

BGIN Meeting Report

Date: TBD

Location: Virtual Meeting

Executive Summary

The transcript captures a detailed discussion on various topics related to digital identity and privacy, including proof of personhood, decentralized identity systems, and post-quantum cryptography. The key points are:

1. **Proof of Personhood:**
 - The concept of separating different types of personal data (e.g., delivery address from purchase intent) to enhance privacy.
 - The use of Siri Knowledge Proofs to maintain data separation and reduce the risk of surveillance.
 2. **Decentralized Identity Systems:**
 - Challenges in operationalizing decentralized identity systems, including user convenience and practical outcomes.
 - The importance of maintaining data separation to prevent multiplicative persona reviews.
 3. **Post-Quantum Cryptography (PQC):**
 - An announcement from Professor Shinichiro Matsuo about a Japanese governance initiative for a PQC competition.
 - Insights into the potential impacts and benefits of adopting post-quantum cryptographic methods.
 4. **Operational Challenges:**
 - The need for practical solutions to ensure user convenience while maintaining strong privacy protections.
 - The role of community input and collaboration in addressing these challenges.
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Key Discussion Points

1. **Proof of Personhood:**
 - **Data Separation:** Maintaining separation between different types of personal data (e.g., delivery address, purchase intent) is crucial for enhancing privacy.

- **Siri Knowledge Proofs:** These proofs allow information to be consumed by both parties without the actual data existing in a single place, reducing surveillance risks.
 - 2. **Decentralized Identity Systems:**
 - **Operationalization Challenges:** Ensuring that decentralized identity systems are user-friendly and convenient is essential for widespread adoption.
 - **Practical Outcomes:** Even with advanced privacy techniques, some personal data (e.g., delivery address) must still be shared to complete transactions.
 - 3. **Post-Quantum Cryptography (PQC):**
 - **Japanese Initiative:** A Japanese governance initiative is launching a PQC competition to advance the development and adoption of post-quantum cryptographic methods.
 - **Potential Impacts:** Post-quantum cryptography can provide robust security against quantum computing threats, ensuring long-term data protection.
 - 4. **Operational Challenges:**
 - **User Convenience:** Balancing strong privacy protections with user convenience is a significant challenge in implementing decentralized identity systems.
 - **Community Collaboration:** Engaging the community and stakeholders is crucial for addressing operational challenges and developing practical solutions.
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Action Items and Next Steps

1. **Further Research on Data Separation Techniques:**
 - Investigate more advanced methods for maintaining data separation to enhance privacy in digital transactions.
2. **User Experience Improvements:**
 - Develop user-friendly interfaces and processes to make decentralized identity systems more accessible and convenient for users.
3. **Participation in PQC Competitions:**
 - Encourage researchers and developers to participate in the Japanese PQC competition to contribute to the advancement of post-quantum cryptographic methods.
4. **Community Engagement:**

- Foster ongoing discussions and collaborations with community members to gather feedback and address operational challenges in implementing privacy-enhancing technologies.

Detailed Session Summary

[Transcript begins with a discussion about the proof of personhood conversation, transitioning into a break and then to a post-quantum cryptography (PQC) discussion.]

Appendix: Full Meeting Transcript (Anonymized)

0.0 --> 1.4: The nature of the word.
3.0 --> 3.6: Okay.
3.8 --> 5.1: We're now recording.
5.6 --> 12.6: I'm going to quickly jump to the previous screen so the recording has it.
13.0 --> 15.2: So this is the BEGIN layer 2 meetup in Sydney.
15.5 --> 20.0: We're bridging standards, governance and technology and sharing the insights from the Tokyo meeting
20.0 --> 20.6: last week.
23.1 --> 27.4: We have three plus one working groups at BEGIN.
27.4 --> 31.9: The identity key access management and privacy working group of which I'm co-chair.
32.5 --> 37.1: The phase or the financial applications and social economies working group.
37.6 --> 39.8: And then the cyber security working group.
40.4 --> 45.2: And the way that we understand this at BEGIN is we've got non-financial applications on
45.2 --> 49.6: one side, financial applications on the other side and cyber security to protect everything
49.6 --> 50.0: in the middle.
51.2 --> 57.4: There is an emergent working group now at BEGIN which is contextually relevant to everything
57.4 --> 62.3: that's accelerating at the moment around agentic AI and a lot of the research that
62.3 --> 68.0: I've been doing recently is to do with agentic AI and you're going to see some of the fruits
68.0 --> 74.9: of that and some of the, I guess, description of where that's going and how BEGIN is placing
74.9 --> 81.5: itself in the agentic future and presenting its knowledge graph to the world.

83.5 --> 92.6: So I'm going to get into the structure of why this research is relevant as a whole to

92.6 --> 99.7: BEGIN and the first session today is on proof of personhood which is extremely relevant

99.7 --> 100.5: to agentic AI.

101.5 --> 109.1: But the research shows that three graphs is stronger than one for proof of personhood.

109.1 --> 112.4: So you could imagine the knowledge graph is the agent.

112.8 --> 117.5: It has access to all the information that BEGIN has either published, all of the transcripts

117.5 --> 124.4: which have then been anonymized and this creates this knowledge graph that we can then share

124.4 --> 125.4: with the community.

126.0 --> 131.6: People can navigate it themselves and I'll show you a fun example of that that I've created

131.6 --> 137.1: a visualization and then share their discoveries and feedback to us which then feeds into our

137.1 --> 143.3: conversations and into our work and our work kind of goes through phases.

143.6 --> 148.4: It starts with discussion, working groups are called and then we progress that research

148.4 --> 156.3: into informing standards organizations like ISO and IEEE and a few others, financial cryptography

156.3 --> 156.7: as well.

158.0 --> 165.6: So the three graph setup or framework which is important for agentic AI is you have the

165.6 --> 167.3: knowledge substrate which I just described.

167.8 --> 170.5: So BEGIN has its own knowledge substrate and community.

171.1 --> 175.3: There is then a promise graph which overlaps with that and that's the human interactions

175.9 --> 183.5: and it's key to understand here that promises are stronger trust building than impositions

184.0 --> 189.2: and the difference between that is if I promise to deliver some research and I then deliver

189.2 --> 193.8: the research, you trust me more because I followed through on that promise.

193.8 --> 198.9: If I just told you I will deliver research and then the research is delivered or not,

199.3 --> 200.7: it's just how it is.

201.2 --> 205.3: So the difference here is an agent can only promise its own behavior.

206.4 --> 214.0: So imagine instead of using I as a person, I had an agent acting on my behalf and I had

214.0 --> 218.6: that agent promise that it's going to deliver research which is the crossover between my

218.6 --> 221.6: knowledge substrate and your work, whatever it is.

222.3 --> 225.3: I then send my agent to go do its thing.

225.5 --> 231.5: It promised to create that overlap between knowledge and then delivers it to you.

231.7 --> 236.5: Your agent will receive that information and because a promise has been fulfilled, it will

236.5 --> 241.1: then make the line between the two agents a little bit more dense.

241.7 --> 245.1: And I'm that guy who joins the group and always talks about graphs.

245.3 --> 247.0: So you might have to get used to that.

247.6 --> 254.9: But graphs are really important, especially in the age like agentic AI era where vertices

254.9 --> 260.1: are the knowledge or the context or the information that an AI consumes.

260.5 --> 265.6: And then the lines between those vertices is actually the promises and what creates

265.6 --> 270.1: weight and gives context between those two vertices or those nodes.

270.1 --> 277.7: So say, for example, let's grab someone from the room, maybe go for it.

278.2 --> 281.3: What is the thing that you've been researching most recently?

283.6 --> 284.7: Could you, Mike?

288.0 --> 290.6: Okay, cool.

291.1 --> 294.9: So settlement finality is like atomic swaps using blockchain, right?

295.7 --> 297.7: Ideal universe would be atomic swaps.

298.6 --> 300.0: Okay, cool.

300.6 --> 300.8: Yeah.

300.9 --> 307.6: So if I then promise to say, I've actually done some research around personhood credentials

307.6 --> 314.7: which could support with that settlement finality to give additional credential weight to, say,

314.7 --> 320.4: for example, the settlement requires like some sort of credit check or some sort of information

320.4 --> 326.8: about the person in order for it to be executed, then what my agent would do is I would promise

326.8 --> 328.5: that my research is relevant to you.

328.7 --> 333.1: You would share some context about your settlement finality work.

333.7 --> 338.5: My agent would consume that information and then it would send back the prompt result

338.5 --> 342.7: to you and you would then obviously consume that and read that.

343.1 --> 347.4: And if you believe that that promise is fulfilled, there is a genuine connection between our

347.4 --> 352.5: two agents where they trust the knowledge that has been shared to be delivered based on the

352.5 --> 353.6: promises that were made.
353.6 --> 360.9: Whereas if I just said, hey, I think my research is relevant to you and I impose my research
360.9 --> 365.3: to you, I'll just send you a paper that I wrote, then yeah, you may read it, but there
365.3 --> 370.2: hasn't been a connection, there hasn't been a line drawn with weight between our two nodes
370.2 --> 372.9: of knowledge, if that makes sense.
373.3 --> 373.4: Yep.
374.8 --> 375.0: Good.
375.1 --> 376.7: Could we make, so the, yeah.
377.2 --> 380.6: That even machines don't make unsolicited advice.
381.8 --> 382.3: Exactly.
382.7 --> 389.9: And this, there is information theory or systems information theory called promise theory by
389.9 --> 397.4: Burgess and I can't remember the second one, that has proven that promises hold more weight
397.4 --> 401.3: than impositions, especially when in agent interactions.
402.3 --> 404.8: Double click on what you had said as an example.
405.5 --> 411.5: If for example, in order for me in an operating level to ensure settlement finality, I required
411.5 --> 417.7: your identity credentials and your agent is the one that's passing it across onto my system.
419.1 --> 425.0: How does this entire promise and trust graph work to solve multiple sources of data that
425.0 --> 429.8: might have to be spooled, like public, private, self-attested, third-party attested?
430.3 --> 435.3: And how is it able to then structure that data and maybe then hand it off to the other
435.3 --> 438.6: vertice or the other agent that's looking to be able to execute this transaction?
439.5 --> 439.7: Yeah.
439.7 --> 443.3: So the way that that would work is both of our agents would make promises.
443.7 --> 448.2: So your promise would then be a positive polarity.
448.3 --> 452.9: So you would have particular directions of these are the requirements for this transaction
452.9 --> 454.4: or this settlement to be made.
455.0 --> 456.7: My agent would receive that.
457.2 --> 463.3: And then it would compose the information based on that feedback, based on its access
463.3 --> 469.1: to my knowledge graph or my information substrate, which is protected through both cryptographic
469.1 --> 472.4: keys and then delegated to through the agent.
472.7 --> 477.1: It would then compose that information, hopefully put it behind a zero-knowledge proof so you
477.1 --> 481.2: don't even need to see the information to know that it

is true and correct, and then
481.2 --> 482.3: send it back to you.
482.7 --> 486.6: And when your agent receives that, it will be like, yeah, this is a zero-knowledge proof
486.6 --> 486.8: call.
487.6 --> 489.0: The correct information is there.
489.5 --> 492.2: And promise has been delivered.
492.8 --> 495.1: We now trust our agents trust each other more.
495.2 --> 496.2: The line is drawn.
496.2 --> 503.0: And the cool thing about that is you then create pathways for the agent navigation.
503.8 --> 508.3: So this is getting a little bit more into the AI theory behind it.
508.6 --> 513.7: But when an AI gets called, it navigates a massive information substrate.
514.7 --> 521.3: So how do we make that navigation more directional and actually path the correct nodes to create
521.3 --> 521.8: efficiency?
521.8 --> 525.4: Well, you add weight to the lines.
525.6 --> 528.3: And these promises add weight to those lines.
528.9 --> 535.3: So it creates an emergent quality, which is this trust graph, which is where knowledge
535.3 --> 539.2: meets promise to then create verifiable relationships.
540.3 --> 546.6: And that emergent graph, part of my research, shows that the trust graph that comes from
546.6 --> 554.2: that is actually really unique per person and means that you don't even need to necessarily
554.2 --> 560.4: share the underlying biometric schema, the passport, these things.
560.5 --> 565.3: You just need to share that there has been a coordination through that trust graph.
565.9 --> 569.4: And the coordinates of that, put that in a zero-knowledge proof, and you get a proof
569.4 --> 570.5: of person credential.
571.4 --> 576.0: So my next question was, are you guys working on creating these standardized schemas or
576.0 --> 579.7: templates for information, which would then be consumed by these agents?
580.0 --> 582.4: But that's not really part of the...
582.4 --> 588.7: Yeah, so that's a little bit out of scope, because that's dictated by the governance
588.7 --> 591.9: of the ecosystem, and every ecosystem has different governance.
592.1 --> 594.2: So this community cares more about knowledge.
594.8 --> 598.0: So we're probably not going to scan people's eyeballs to let them in.
598.6 --> 611.1: We had WorldID present at Begin, and it could be a future

re, but hopefully we have those ecosystems

611.8 --> 616.5: create their own governance, and part of that ecosystem's governance, for example, Begin,

616.9 --> 621.7: we more care about knowledge sharing and the multi-stakeholder dialogue and maybe a credential

621.7 --> 622.4: here and there.

622.5 --> 625.7: But it is open to anyone.

626.3 --> 633.0: So realistically, we care about people contributing as the personhood of our ecosystem.

633.8 --> 635.9: And you could imagine different ecosystems.

636.2 --> 644.1: So a university might require a few more credentials, say, for example, you're in a course or you've

644.1 --> 646.5: received a degree or something like that.

648.0 --> 654.2: But yeah, creating the templates for the personhood ecosystem is a little bit out of scope.

655.0 --> 656.9: However, we do need to start somewhere.

657.5 --> 663.6: So the ecosystems that apply this research first, obviously, we're going to support them

663.6 --> 666.9: in proliferating that system, that substrate.

666.9 --> 667.1: Great.

678.9 --> 684.5: Past behavior doesn't necessarily affect future behavior, so that trust may be established

684.5 --> 689.0: as maybe a way of getting a comment.

689.5 --> 689.7: Yep.

689.9 --> 694.1: You know, the trust is established, but then it's systematically broken.

694.1 --> 701.6: When it is broken, does it sever the full thick cord or does it only sever part of the

701.6 --> 701.9: cord?

701.9 --> 703.0: That's a fantastic question.

703.6 --> 710.9: So the other great thing about promise theory is it's not a collapse of trust, it's a degradation

710.9 --> 711.7: of trust.

712.4 --> 720.9: So if you break a particular promise related to some sort of scam situation that happens,

720.9 --> 726.0: it's just within that promise that trust is broken and the line has less weight.

726.6 --> 730.8: However, it doesn't fully collapse the connection between the nodes.

731.4 --> 737.1: And this is important because, yeah, agents will act outside of their instructive behavior

737.6 --> 742.6: and we need to work out where that happened in retrospect.

743.1 --> 750.3: So if you align those promises from the start, you can actually pinpoint at which promise

750.3 --> 755.4: did the agent not fulfill because the other one is observing its actions.

756.6 --> 764.9: And then that creates that incremental degradation of trust rather than complete collapse of
764.9 --> 765.2: trust.
769.2 --> 773.5: And, yeah, the final one here is the temporal proof.
773.7 --> 779.2: And that's why blockchain is relevant, because blockchain is that deviation chain that actually
779.2 --> 780.4: attaches itself to time.
782.2 --> 787.0: So you put a zero knowledge proof on a blockchain and you can work out when these things were
787.0 --> 787.9: executed or shared.
792.7 --> 797.4: So we've kind of already jumped right into this and thank you for being involved.
797.7 --> 800.1: This has been brilliant and what we're looking for.
801.0 --> 803.2: So part of the research is the agent duality.
803.6 --> 808.1: And I've come up with more like mimetic names for these.
808.1 --> 809.7: So the swordsman and the mage.
809.8 --> 812.8: But you could understand the swordsman to be the cryptographic keys.
813.0 --> 815.3: So your protection, it's the thing that sets boundaries.
815.7 --> 818.4: It's the thing that cuts surveillance and observation.
819.2 --> 821.6: And then the mage is your delegation or agency.
822.5 --> 824.7: So it's the thing that projects information.
825.2 --> 829.3: It accepts information in and that information may have a prompt attack.
829.4 --> 831.4: It may have some sort of issue.
831.8 --> 837.2: So we shouldn't accept information into our cryptography and into our private keys.
837.2 --> 838.3: We need to have a layer.
838.8 --> 844.4: So the mage accepts foreign inputs and it delegates information.
844.6 --> 846.1: It does all of the knowledge stuff.
846.2 --> 853.2: And then it sends it through a trust spanning layer to swordsman, which makes the commitments
853.2 --> 853.8: on chain.
854.0 --> 857.8: Does the zero knowledge proofs, controls your keys, sends the transactions.
858.4 --> 864.2: And this separation is really interesting because when you do create a mathematical separation
864.2 --> 867.4: between these two agents, they still need to create a language between them.
868.0 --> 872.2: And that language is what I'm most interested in because that is an emergent property.
872.7 --> 877.4: And what I predict for the emergence to be is two qualities.
877.5 --> 878.3: One is reflection.
878.7 --> 883.4: And reflection is the ability to see when something happened at a particular point in

883.4 --> 887.6: time and reflect upon that and see that over time.
888.0 --> 891.9: And then the other is connection, which we all do naturally.
891.9 --> 896.2: But when we do share knowledge with each other, there is a connection made in the graph.
896.3 --> 899.0: And that's what we were talking about between the agents making promises.
900.0 --> 903.4: So the connection quality is we've shared knowledge.
903.9 --> 911.5: And these two agents are now trusting each other to share that specific knowledge
911.5 --> 911.8: set.
913.2 --> 914.1: Yeah, go ahead.
918.4 --> 920.6: Within the quantum environment as well.
920.6 --> 923.2: So this is part of why it's important.
924.3 --> 931.2: So that separation actually preserves the first person to like a quantum like attack
931.2 --> 937.2: in a way where because there is a mathematical separation between these two knowledge substrates,
937.5 --> 942.5: if something attacks, it becomes additive, not multiplicative.
943.0 --> 949.3: So if you're observing one surface and you see all of that information with a quantum
949.3 --> 954.8: computer, you become fully reconstructable and it's multiplicative, that information
954.8 --> 958.1: to like the adversarial observer.
958.6 --> 963.4: Whereas if you have these systems separate and one of them has the cryptographic keys
963.4 --> 968.1: and one of them has the information and the trust tasks, then only one of those surfaces
968.1 --> 974.9: gets compromised, which means that you can never be fully deep faked or fully reconstructed
974.9 --> 978.2: in the case of like a quantum attack.
980.9 --> 985.6: So yeah, one is to protect and the other is to project information.
986.3 --> 990.0: And the fun term that I've come up with is swordsman and mage.
992.8 --> 998.6: And the emoji inscription at the bottom is the spell.
998.6 --> 1006.0: So if my agent was to send you this spell and your agent has consumed my knowledge, it
1006.0 --> 1012.0: would be able to episodically look up all of the information that I've shared about
1012.0 --> 1018.9: agent duality using only 10 tokens rather than having to learn the entire Git repo again
1018.9 --> 1021.6: because this equation here is a compression key.
1021.6 --> 1028.1: So it's the compressed understanding of the agent duality equation.
1031.3 --> 1036.2: So I went through this in some good detail.

1036.6 --> 1038.3: You have the three graphs, one trajectory.
1038.7 --> 1042.9: So you are on one trajectory through these three graphs and it is the overlap within
1042.9 --> 1047.7: these graphs which creates a personhood credential depending on the knowledge substrate.
1048.4 --> 1050.3: And yeah, Begin has a knowledge substrate.
1050.9 --> 1052.2: Biometrics have a knowledge substrate.
1052.6 --> 1057.8: These are all just information systems that you then access and control.
1058.4 --> 1063.7: But in order for you to sort of maintain your sovereignty over that,
1063.9 --> 1066.5: we create this three graph system and the overlap.
1068.1 --> 1072.5: And the emergent trust graph is where we're going.
1076.1 --> 1080.8: Okay, so I'm going to quickly show.
1082.0 --> 1089.9: This is the agent privacy research and you can see here this is a topological
1089.9 --> 1093.2: knowledge graph of all of the research and the white papers and
1093.2 --> 1097.4: the living documentation that I've created for agent privacy.
1098.0 --> 1102.4: And we're going to have like a fun little example experiment here
1103.1 --> 1107.6: where I'm going to pick one of these.
1113.0 --> 1115.7: The Australian internet goes far here.
1120.3 --> 1124.2: But we'll start here because this is the master inscription that I just shared.
1124.8 --> 1126.8: And let's start a waypoint.
1129.3 --> 1133.1: And if we could have, there's a mic over there as well.
1133.6 --> 1134.5: I'm going to call on you.
1134.9 --> 1135.8: Nice to meet you.
1136.9 --> 1142.4: If you could pick one of these three options here of us navigating
1142.4 --> 1147.0: the knowledge substrate that I'm making publicly aware to the world.
1150.5 --> 1151.6: Might need a turn.
1155.1 --> 1156.0: Hopefully it works.
1156.0 --> 1157.5: Otherwise we can just go with the one.
1158.2 --> 1158.6: Yeah, yeah.
1162.3 --> 1162.7: Sorry.
1169.5 --> 1170.7: Might have just started work.
1170.8 --> 1171.3: Okay, yeah, cool.
1173.2 --> 1174.8: I'll take the second one.
1175.2 --> 1176.3: So conditional independence?
1177.3 --> 1179.0: All right, we'll add that to the path.
1180.9 --> 1182.8: Okay, now, mic over there.
1183.0 --> 1184.8: Let's pick one from the list here.

1186.0 --> 1189.8: How are you going to navigate the arena?
1190.4 --> 1192.1: If you could pick one from the list.
1194.8 --> 1195.7: Separation theorem?
1198.5 --> 1199.9: We'll add that to the path.
1204.8 --> 1207.5: Okay, I'm going to add the research paper.
1209.9 --> 1213.9: Now could we get another choice?
1214.4 --> 1215.1: Just call it out.
1221.8 --> 1223.7: Okay, that's already in the path.
1224.8 --> 1229.7: So we'll add the swordsman agent.
1230.6 --> 1233.8: I'm also going to add my terms because this is relevant and
1233.8 --> 1234.9: I'll include this later.
1235.3 --> 1236.9: It's how we get the promise graph.
1237.5 --> 1240.0: I'm just going to add a few of the navigations here.
1242.4 --> 1244.6: And we'll finish off.
1254.1 --> 1259.9: So this is, what I've done here is I've consumed all of
1259.9 --> 1265.7: the research in a Git repo and then turned it into a graph.
1266.2 --> 1267.6: And this is a knowledge graph.
1268.3 --> 1274.4: And I guess, to put it in a regulatory context, let's search.
1279.7 --> 1283.8: Some of these nodes light up when you search regulatory.
1284.3 --> 1288.2: So this one, for example, it's policy governance.
1288.6 --> 1289.7: It's already in the path.
1291.2 --> 1292.0: We'll pick this one.
1293.5 --> 1295.0: The gap is in the path.
1296.0 --> 1297.3: We'll add this one in.
1298.8 --> 1300.5: Ethereum is already in the path.
1301.4 --> 1302.9: So we'll add agent identity.
1303.8 --> 1305.3: Okay, I'm going to close the portal here.
1306.6 --> 1309.1: And this is me creating a unique string.
1311.3 --> 1315.6: So this is my unique compression of my navigation through
1316.2 --> 1317.9: Mitchell's research knowledge graph.
1318.9 --> 1320.5: This is a reflect quality.
1333.2 --> 1335.3: So this adds weight and
1335.3 --> 1339.0: information to this particular trajectory that we took through the knowledge graph.
1342.4 --> 1345.4: And what's unique here is it creates a constellation.
1347.8 --> 1349.9: And if you were to evoke the constellation,
1350.4 --> 1356.3: it gives you this particular spell here, which is a compression of the anti-mirror.
1356.8 --> 1360.9: And the anti-mirror is, you cannot be fully reconstructed if you have
1360.9 --> 1364.7: conditional independence and separation between two i

nformation systems.

1365.1 --> 1366.5: And this is proven, this is a theorem.

1367.2 --> 1370.6: Claude Shannon came up with it in 1970.

1372.2 --> 1373.8: So yeah, Claude, namesake.

1374.0 --> 1376.0: He was an information theorist.

1378.1 --> 1384.9: And I guess the example that I wanted to share here is what we did here today,

1384.9 --> 1389.6: navigating a particular knowledge substrate, is extremely unique.

1390.5 --> 1394.5: Every person is going to have a different path through the knowledge substrate based

1394.5 --> 1400.4: on their context, their information, where they are at a particular space and time.

1400.8 --> 1406.9: So if we were able to encode this as a proof of personhood credential that I,

1408.1 --> 1411.3: agent privacy or begin, then shared with you,

1411.3 --> 1414.8: saying that you went through this knowledge substrate and

1414.8 --> 1420.3: created this specific constellation, we did a proof of personhood.

1420.5 --> 1423.4: I understand that you're a person that went through the knowledge substrate.

1423.9 --> 1428.3: An AI agent is never gonna be able to do this because an AI agent or

1428.3 --> 1433.2: an LLM will always find the path of least resistance to the query or the prompt.

1434.1 --> 1437.8: So the fact that you're a human making choices on this path

1438.4 --> 1439.8: creates unique personhood.

1439.8 --> 1442.6: And if we're able to then scale this,

1442.7 --> 1446.4: you could imagine many different types of knowledge substrates, for example.

1447.1 --> 1454.2: So it gives opportunity to issue proofs as coordinates,

1454.3 --> 1458.5: rather than issuing them as the underlying data or

1458.5 --> 1461.2: information or the personally identifiable information.

1462.2 --> 1468.6: And if information never exists, it's far easier to protect than if it exists.

1468.6 --> 1471.4: So if someone gives their credential to,

1471.6 --> 1475.8: this is the bartender problem, to get into a bar to prove they're over 18.

1476.3 --> 1481.1: Well, the bartender observed your passport or driver's license.

1481.4 --> 1485.7: They get to observe your address, your date of birth,

1485.8 --> 1488.0: all of these things just to check that you're over 18.

1489.1 --> 1491.7: So this would provide a solution to that.

1491.8 --> 1494.1: I would have a particular personal knowledge substrate.
1494.1 --> 1498.2: I would then be able to compose a specific constellation saying that,
1498.3 --> 1503.2: I've had a 21st birthday, I added that, there's a photo of it.
1503.5 --> 1509.3: I then compose a proof saying that I'm over 18, and these people have observed that.
1509.8 --> 1513.9: Some sort of educational institution has issued a credential
1514.5 --> 1516.4: that also says that I'm over 18.
1517.1 --> 1520.5: I'm the one composing that proof for the bartender.
1520.5 --> 1525.5: Bartender never sees the underlying information that created that range proof
1525.5 --> 1528.6: that I am over 18 and issues it.
1529.4 --> 1532.2: Bartender observes the zero knowledge proof and lets you in.
1534.1 --> 1534.7: I'm gonna pause here.
1535.8 --> 1539.6: Any sort of questions or thoughts or,
1540.4 --> 1543.1: I'm most interested in how this reflects to your own work.
1543.4 --> 1546.9: And if this spurs any thoughts on how you could
1547.4 --> 1549.6: evolve the things that you're doing and working on.
1552.1 --> 1552.8: Any thoughts?
1564.0 --> 1566.0: Possibly more questions on this.
1566.6 --> 1568.1: Keep your interest active.
1569.2 --> 1571.0: So I can let a panelist at this,
1571.1 --> 1574.8: let them at this develop things to more show the ethnic
1575.1 --> 1580.3: that you don't want, how would this standard be used?
1581.1 --> 1583.8: Yeah, so there are standards in.
1587.2 --> 1587.8: Yeah.
1588.3 --> 1591.9: So if you are using a standard in your work, for example,
1592.1 --> 1592.9: we'll use your example.
1593.0 --> 1595.0: It's a good one for the settlement.
1595.6 --> 1597.3: So if you're using a particular standard
1598.2 --> 1601.4: to execute the transaction for your work,
1601.7 --> 1603.4: that would exist within here.
1603.4 --> 1606.9: And then if an agent is calling that standard or data
1606.9 --> 1610.7: structure, then it will navigate to that point
1610.7 --> 1612.4: in the knowledge graph in order to get
1612.4 --> 1614.8: the right information to then execute on the standard.
1615.4 --> 1617.9: So if you, as a person, would like
1617.9 --> 1620.0: to set up your own agent workflow,
1620.9 --> 1623.4: you draw a constellation that has
1623.4 --> 1624.8: all of the correct standards.

1625.1 --> 1632.3: You then save it, and you can share that to an agent.
1632.3 --> 1635.4: And then the agent has an instructional workflow
1635.4 --> 1637.1: of how to navigate your knowledge
1637.1 --> 1639.9: graph in order to execute that particular standard.
1640.5 --> 1642.0: So that's one way.
1642.5 --> 1646.0: The other way is this also accelerates
1646.0 --> 1650.4: the adoption of standards with agentic AI.
1650.7 --> 1653.7: So the fact that standards exist within this knowledge
1653.7 --> 1657.5: substrate means that anyone who uses the agent privacy
1657.5 --> 1661.7: protocol automatically uses the IEEE 7018,
1661.7 --> 1664.8: which is my terms, that I'm going
1664.8 --> 1667.0: to explain to you guys in a sec.
1669.2 --> 1670.6: Mitch, I've got a question.
1671.2 --> 1671.6: Yeah, go ahead.
1671.8 --> 1673.6: Just for context, my name is Shankar.
1673.9 --> 1676.8: I work predominantly in the financial services space
1676.8 --> 1678.8: in governance and risk management.
1680.2 --> 1683.7: So the proof of personhood is a really interesting one,
1683.9 --> 1686.1: because the regulator, again, this is a regulatory angle,
1686.9 --> 1690.5: they're very interested in things like know your customer,
1690.5 --> 1692.0: know your business, et cetera.
1692.2 --> 1696.5: So how do you see this making that jump
1696.5 --> 1699.0: to being able to satisfy a regulator
1699.0 --> 1701.7: that a financial institution has done
1701.7 --> 1704.1: enough to validate a person?
1704.8 --> 1708.3: Yeah, I'm going to quickly jump back to our presentation,
1709.4 --> 1711.9: because this is a good lead in there.
1714.6 --> 1718.8: So in the financial and regulatory space,
1718.8 --> 1720.8: there are particular KYC requirements.
1721.4 --> 1724.2: And those KYC requirements are information
1724.2 --> 1725.5: or knowledge disclosures.
1726.2 --> 1730.2: And those knowledge disclosures, if put behind a zero
knowledge
1730.2 --> 1733.3: proof, is actually kind of hard to verify
1733.3 --> 1736.6: unless you have the ability to query that proof
1737.1 --> 1738.2: and unwrap it yourself.
1738.9 --> 1742.0: So I would make a promise to you to be like,
1742.1 --> 1746.6: you have the ability to navigate this particular path
1747.3 --> 1753.1: and unwrap the proof to show that I have shared
1753.1 --> 1758.9: a biometric KYC information in order to prove that I'm a human

1758.9 --> 1760.9: if you use World ID, for example.
1762.4 --> 1767.3: Another way that it's relevant, which we were explaining before,
1767.6 --> 1771.4: is it's not only the KYC information,
1771.6 --> 1774.7: it could also include my understanding
1775.3 --> 1776.0: of the transaction.
1777.0 --> 1779.3: So if I had a particular understanding
1779.3 --> 1783.0: of the transaction, I could then compose that path,
1783.3 --> 1785.5: that constellation, to be like, yeah, this
1785.5 --> 1786.9: is like a 10 ETH trade.
1787.4 --> 1789.1: I would like to do it with this broker.
1789.9 --> 1791.1: I encode that.
1791.3 --> 1792.7: I prompt that into my agent.
1792.9 --> 1797.6: It navigates the path to ensure that your request for KYC
1797.6 --> 1798.2: information.
1798.7 --> 1800.3: I think this goes past KYC.
1800.4 --> 1801.7: This is better than KYC.
1801.7 --> 1803.1: So I just want to add that there.
1804.5 --> 1808.5: Your request for information is then received by my agent.
1808.6 --> 1812.4: My agent then composes the bare minimum to reach that request.
1813.6 --> 1817.0: It may navigate to my biometric identity
1817.0 --> 1820.3: if you need some level of personhood that adds biometrics.
1820.5 --> 1823.4: It may navigate to my broker, accountant,
1824.1 --> 1826.6: like proof of wholesale investor certificate
1826.6 --> 1830.3: or something like that, and connect all of those dots,
1830.3 --> 1832.3: compose it as a zero-knowledge proof,
1832.8 --> 1834.7: and then selectively disclose that to you.
1835.4 --> 1838.4: And what's interesting here is you send that request
1838.4 --> 1840.8: to the mage agent, which has access
1840.8 --> 1842.1: to all of my knowledge substrate.
1842.5 --> 1843.7: But you could attack me.
1844.1 --> 1846.4: You could try and add a prompt injection or something
1846.4 --> 1848.0: like that when everything's agentic.
1848.4 --> 1851.7: So that mage agent is actually protecting me
1851.7 --> 1854.8: because the thing that sends the proof is the swordsman.
1855.0 --> 1856.8: So the mage composes that information.
1857.5 --> 1859.0: It then wraps it up.
1859.0 --> 1862.6: And the swordsman agent then sends that proof to you.
1863.0 --> 1865.7: And that means that there is, again,
1865.8 --> 1867.9: a separation in those information systems.
1868.7 --> 1872.4: So I then make a discrete measurement on a blockchain
1872.4 --> 1875.5: to say this is the proof that I have completed

1875.5 --> 1877.1: that sort of KYC requirement
1877.1 --> 1878.7: for this transaction to be executed.
1881.7 --> 1883.9: Okay, so there, yeah, yeah, go on.
1883.9 --> 1884.0: Go on.
1885.1 --> 1885.6: Mitch.
1887.2 --> 1888.9: There's been interest in privacy.
1890.6 --> 1892.6: And is the crux of this,
1892.7 --> 1894.2: when you say the minimum requirement,
1894.4 --> 1895.7: and then you add more if necessary,
1896.3 --> 1899.2: so is that how we're becoming privacy-assured?
1899.7 --> 1899.8: Yeah.
1901.9 --> 1904.4: Many of my cyber friends say that this,
1904.7 --> 1908.7: the ID component, authenticating that you are who you
are,
1909.1 --> 1911.4: is the crux of the process.
1912.0 --> 1912.9: Great, you go first.
1914.3 --> 1916.0: My handle is Privacy Mage.
1916.4 --> 1918.0: I'm all about privacy.
1918.4 --> 1920.9: I think I've actually, part of the research,
1921.1 --> 1924.3: I've created what's called the privacy is value equat
ion,
1924.8 --> 1928.4: where I'm trying to prove that it is actually privacy
1928.4 --> 1929.8: that creates value.
1931.2 --> 1935.4: And it is, the moment that you disclose that informat
ion
1935.4 --> 1938.4: is when the value starts being observed
1938.4 --> 1941.4: and losing its value.
1941.9 --> 1943.4: And in surveillance economies,
1944.5 --> 1947.4: that is completely extracted the moment it is created.
1948.2 --> 1950.9: However, in this setup or system,
1951.8 --> 1954.1: you have self-sovereign control
1954.1 --> 1955.6: of that data and information,
1955.7 --> 1957.3: and you're the one who draws the lines.
1957.4 --> 1959.7: You're the one who makes the promises between the age
nts
1959.7 --> 1962.9: that actually add weight to that information and data.
1963.2 --> 1965.5: So then the value of that information
1965.5 --> 1967.0: is exponentially higher.
1968.3 --> 1971.1: And if you selectively disclose that information
1971.1 --> 1972.7: and put it behind a zero-knowledge proof,
1972.7 --> 1975.9: that doesn't exist to the surveillance economy.
1976.4 --> 1978.6: So you're maintaining the value in yourself.
1978.8 --> 1981.1: So this creates a data dignity model.
1981.9 --> 1984.1: And part of my research actually uses,
1984.6 --> 1985.9: there's a book called Polarity
1985.9 --> 1988.4: that I suggest you check out
1989.9 --> 1992.6: that goes into this concept of data dignity,

1992.8 --> 1993.8: which is important here.
1994.7 --> 1996.3: So yeah, privacy nerds love this.
1997.1 --> 1997.7: Yeah.
1998.4 --> 1999.2: Sounds really good.
1999.2 --> 2004.4: Do you think you're overly reliant on that idea
2004.4 --> 2008.7: that an AI agent follows the path of least resistance?
2009.7 --> 2013.0: That sort of works in, sorry, I've started.
2013.1 --> 2013.9: No, no, no, this is good.
2014.4 --> 2018.1: If you're right to see why in a particular path
2018.1 --> 2020.9: or multiple paths, sort of like order,
2021.5 --> 2024.3: but is that true?
2024.6 --> 2024.8: Does it?
2025.0 --> 2026.7: Yeah, so AI is probabilistic.
2026.8 --> 2029.0: So you're right to like call back on that.
2029.0 --> 2031.4: Some AI is deterministic models,
2031.6 --> 2033.2: which follow that path of least resistance.
2033.8 --> 2035.3: But the fact that it's probabilistic
2035.3 --> 2037.8: actually adds some randomness to the path.
2043.5 --> 2046.7: Yeah, so I expect for that to be an attack.
2047.2 --> 2053.7: But the way that I've sort of conceptually understood
that
2053.7 --> 2056.0: is the more times you do this,
2056.1 --> 2058.5: the more unique that path is going to be.
2058.5 --> 2059.7: Like uniquely human.
2060.2 --> 2062.5: So we did one constellation today
2063.2 --> 2065.7: that's probably very reconstructable
2065.7 --> 2068.7: by any AI that observed this conversation.
2070.0 --> 2072.1: But if we did this again every day,
2072.2 --> 2074.4: and we did this as part of a research group
2074.4 --> 2079.2: that we're exploring and created those like coordinates
2079.2 --> 2080.5: every step of the way,
2080.7 --> 2084.5: it's actually the map between those constellations
2084.5 --> 2087.5: that creates like a unique path
2087.5 --> 2089.1: that an AI couldn't reconstruct.
2091.6 --> 2095.6: I have a somewhat unrelated question.
2097.5 --> 2101.4: When I was looking at the swordsman mage thing,
2101.7 --> 2104.7: it reminded me of a friend of mine's book,
2104.9 --> 2106.7: From Hoodies to Suits.
2108.0 --> 2110.1: In the sense that one of the things
2110.1 --> 2111.9: that she kind of talks about is the difficulties
2111.9 --> 2113.5: that the crypto industry had
2114.0 --> 2117.1: because of kind of like how it presented itself
2117.1 --> 2119.2: or taking like things that existed
2119.2 --> 2121.6: and renaming it new things and creating this confusion,
n,

2121.7 --> 2123.7: which I'm not saying like it's confusing,
2124.5 --> 2126.8: but as like a standard setting organization,
2127.2 --> 2129.8: do you think that as much as I think it's fun,
2130.1 --> 2132.3: like I'm a little confused
2132.3 --> 2134.0: between the swordsman and the mage
2134.6 --> 2138.7: in the sense that like, do you think it takes away from
2138.7 --> 2142.0: or that in presenting this as like a standard setting
body,
2142.0 --> 2143.9: that if we have like these fun names,
2144.7 --> 2146.9: like when you first sent it, I thought it was a game.
2147.1 --> 2148.5: And don't get me wrong, gamification.
2148.7 --> 2149.6: It kind of is a game though.
2149.9 --> 2151.3: Yeah, but and that's the thing is,
2151.4 --> 2153.7: if it's a matter of gamification, that's fair.
2153.7 --> 2157.2: I've had this comment and feedback actually a fair amount.
2157.6 --> 2158.3: Someone was just like,
2158.4 --> 2160.4: looks like you're just creating a Dungeons and Dragons thing
2160.4 --> 2161.8: and those sorts of things.
2163.2 --> 2164.3: I agree with you.
2164.5 --> 2168.8: I think part of it is that the word swordsman to me
2168.8 --> 2170.0: has more weight to it.
2170.0 --> 2172.3: And the word mage has more weight to it.
2172.9 --> 2176.1: When you start adding it to AI,
2176.3 --> 2181.6: I think they add, those words mean more in this context.
2182.3 --> 2182.6: To what?
2182.7 --> 2183.6: Like in the sense of like.
2183.7 --> 2188.1: Protect and project is another, that's the two,
2188.4 --> 2190.0: that's the duality as well,
2190.3 --> 2194.3: or privacy and agency is also the duality.
2194.9 --> 2197.1: So if you wanna be more serious about it,
2197.5 --> 2199.6: you can create the separation here
2199.6 --> 2204.3: where the top is agency and the bottom is privacy
2204.3 --> 2207.6: or protect and project.
2209.2 --> 2211.4: So yeah, I've had that feedback
2211.4 --> 2214.2: and I'm thinking seriously about it,
2214.3 --> 2218.3: but part of the beauty of this system
2218.8 --> 2223.0: is that what LLMs are really good at is two things.
2223.1 --> 2224.7: One is compressing information
2224.7 --> 2227.4: and the other is reconstructing that information
2227.4 --> 2229.0: within a particular context.
2229.0 --> 2231.6: So your context or my context or whatever.
2232.4 --> 2237.6: And that process requires persona or personality
2238.2 --> 2242.2: in order to add more specific outcomes

2242.2 --> 2244.9: to the reconstruction of that information.
2245.7 --> 2249.2: So that persona being, I'm a mage casting spells
2249.2 --> 2251.0: trying to share knowledge and information,
2251.9 --> 2254.2: or I'm a swordsman creating boundaries
2254.2 --> 2255.5: and cutting surveillance,
2256.2 --> 2260.8: actually helps the AI with getting more creative outcomes
2261.6 --> 2263.7: for the compression of that information
2264.3 --> 2267.6: for me to then send you just a emoji string or a proverb
2268.4 --> 2271.1: of like the full reconstruction of the information.
2273.4 --> 2276.5: So yeah, it's a great comment.
2276.8 --> 2277.9: I've received the comment a lot.
2278.8 --> 2279.6: I think-
2281.0 --> 2282.7: Just to be clear from a confusion standpoint,
2283.8 --> 2286.9: the swordsman and the mage are just different names
2286.9 --> 2288.9: for sovereign wallet and sovereign agent.
2289.2 --> 2289.3: Correct.
2289.4 --> 2291.0: So you could just literally-
2291.0 --> 2291.4: You could, yep.
2291.5 --> 2292.0: Okay, all right.
2292.3 --> 2293.2: Those are the equivalent.
2293.4 --> 2297.1: It's the agent privacy expression of this duality,
2297.3 --> 2299.8: which is essential for us to maintain our sovereignty
2299.8 --> 2304.1: when delegating knowledge and keys to agents.
2305.0 --> 2307.8: And my suggestion is this bottom sovereign wallet one
2307.8 --> 2310.6: is probably really, really late,
2310.6 --> 2313.9: like an AI getting access to that information
2313.9 --> 2318.3: is hopefully happens far later than the top
2318.9 --> 2321.4: where an AI starts doing trust tasks for us.
2321.6 --> 2324.3: What we did with the constellation is a trust task.
2325.4 --> 2328.7: All the UI experiences where I delegate an agent
2328.7 --> 2329.9: to do my travel booking.
2330.5 --> 2332.7: So the top happens right now already
2332.7 --> 2335.3: and all of the AI agent stuff in the market
2335.3 --> 2336.7: is mostly all the top.
2337.6 --> 2338.1: In the bottom.
2338.8 --> 2341.2: This is the trust task, the sovereign agent.
2341.7 --> 2342.6: Yeah, the sovereign agent.
2344.2 --> 2346.5: So the trust spanning protocol is the middle.
2347.1 --> 2350.0: The bottom is verifiable credentials, digital assets,
2350.2 --> 2352.4: self-certifying and cryptographic keys.
2352.9 --> 2355.8: Not the left side, I'm with-
2355.8 --> 2360.2: Yeah, so this like stack, the human stack.
2360.8 --> 2363.7: And the human stack is modeled off the IP,
2364.1 --> 2364.9: like the internet stack.
2364.9 --> 2370.0: So the top in like IP is the information spanning pro

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2370.3 --> 2373.0: The top is software and the bottom is hardware.
2373.4 --> 2376.8: So this mirrors that, but this is the human stack,
2376.9 --> 2380.2: not the machine stack.
2381.6 --> 2382.1: Yeah, sorry.
2384.8 --> 2387.7: Actually, I probably, it should be swapped.
2388.0 --> 2389.8: That's a good note.
2390.0 --> 2392.3: So the sovereign should be at the bottom on the right
side
2392.3 --> 2394.0: and the main should be at the top to reflect.
2394.0 --> 2395.3: But yeah.
2395.6 --> 2395.9: That's okay.
2396.0 --> 2397.9: Sorry, I just was looking to the right of your head.
2398.7 --> 2400.8: So that's why I was getting back on track.
2402.7 --> 2404.4: Amanda, your point's a really good one
2404.4 --> 2406.6: about the words we use and the semantics
2406.6 --> 2409.4: and the corporate speak is very deep.
2410.0 --> 2412.0: Part of the, I'm in house counsel
2412.5 --> 2415.4: and part of my job is to try and understand the techn
ical
2415.4 --> 2417.7: and then explain it to board language,
2417.9 --> 2421.1: in board language for people who have the purse strin
gs
2421.1 --> 2423.1: to say, let's invest in this.
2423.1 --> 2425.8: So if we could use some sort of consistent,
2427.7 --> 2430.2: when I say this, I mean this.
2431.4 --> 2433.4: So probably something that's an acronym
2434.3 --> 2435.8: as the word protocol.
2436.1 --> 2439.3: I know there's certain language that certain groups u
se
2439.3 --> 2441.8: that if we could speak a.
2442.3 --> 2442.5: Yeah.
2447.1 --> 2449.0: I think you make, yeah.
2449.1 --> 2450.4: And I think you make two points,
2450.9 --> 2454.0: even a better point than I was making,
2454.6 --> 2456.7: which is also like not just the words,
2456.9 --> 2459.6: but the public facing words
2460.6 --> 2462.9: versus whatever it is that you actually use
2462.9 --> 2463.9: on the backend, right?
2464.0 --> 2466.1: Because like having been like chief of legal affairs
2466.1 --> 2469.0: in a company, I know that there's the marketing side
2469.0 --> 2470.7: that we show to the outside of the world.
2471.1 --> 2472.7: And then there's what the engineers tell me.
2472.8 --> 2475.7: And there's an aspect to this, 100% agree.
2476.3 --> 2479.7: If what you're saying or what I'm hearing you say is
2479.7 --> 2481.4: that the swordsman, the mage,
2481.7 --> 2484.7: like even though they may seem like dungeons

2484.7 --> 2485.9: and dragons are silly,
2486.1 --> 2489.3: there is actually an engineering value to use.
2490.3 --> 2493.7: But at the same time, like nobody needs to know that,
right?
2493.7 --> 2495.8: Like the outside world does not need to see that.
2496.1 --> 2498.3: And it's not even a matter of need.
2498.4 --> 2501.8: It goes back to like the harm of what happens
2501.8 --> 2505.2: if you turn people off because humans are sheep
2505.2 --> 2507.5: and they expect like a certain thing in their setting
2507.5 --> 2509.2: in the way that they expect it
2509.2 --> 2512.7: versus like, if you're talking behind the scenes
2512.7 --> 2515.1: about, well, this is what really happens, right?
2515.4 --> 2517.9: Then I think there's almost like two layers of,
2518.7 --> 2521.7: this is what we say on the outside, right?
2521.7 --> 2525.1: For everybody to get it versus this is actually how w
e do it
2525.1 --> 2527.0: so that the code and everything works
2527.0 --> 2528.6: and it's actually more useful.
2529.2 --> 2531.7: And it's not to say that that one's wrong.
2531.8 --> 2533.2: It's just to say that like,
2534.1 --> 2535.8: they have different uses and different.
2536.8 --> 2538.2: The word I use is expressions.
2539.2 --> 2544.4: There are different expressions of the underlying arc
hitecture,
2544.9 --> 2548.3: protocol, primitives and standards.
2549.0 --> 2553.8: And the swordsman and the mage is the colloquial expr
ession
2553.8 --> 2558.0: of the agent duality between sovereign agent
2558.0 --> 2565.2: and sovereign wallet is a more corporate standard spe
ak
2565.2 --> 2566.9: of that expression.
2568.6 --> 2571.8: Part of one of the fun things
2571.8 --> 2573.7: that I've done with this research
2574.2 --> 2578.2: and it does go as low as like maths
2578.2 --> 2580.9: in terms of compression and protocol language.
2581.5 --> 2584.8: And that sits within a white paper
2584.8 --> 2587.1: and a theoretical paper as well.
2588.6 --> 2590.7: But I've actually consumed all of that.
2590.9 --> 2594.6: And if we remember the spell web, that beautiful grap
h,
2595.2 --> 2599.9: I've reconstructed the entire protocol language archi
tecture
2600.5 --> 2601.9: as a story as well.
2602.3 --> 2603.6: And I call them spell books.
2604.3 --> 2605.7: And there are five.
2606.0 --> 2608.5: The first is the story spell book.
2608.6 --> 2609.5: So that's my journey.

2609.9 --> 2611.3: So it started with 10 acts.
2611.8 --> 2615.4: And then over the time when experts,
2615.5 --> 2617.6: different people have observed my work
2617.6 --> 2620.5: and said, this is me naming your work.
2620.9 --> 2622.0: And this is relevant.
2622.5 --> 2626.1: It then creates another story, another act in the spell book.
2626.7 --> 2630.7: And part of the reason for that is it's my belief
2630.7 --> 2633.1: that in order to train inference of an agent,
2633.4 --> 2635.0: you have to send it on a journey.
2635.7 --> 2637.2: You don't give it the destination.
2638.0 --> 2642.1: It has to go through a path of this is what this means.
2642.9 --> 2645.6: And then we go to the next part of the story.
2645.7 --> 2647.0: And this is what that means.
2647.1 --> 2650.2: And I actually choose depending on the outcome of the agent,
2650.9 --> 2652.2: what the story is.
2652.7 --> 2654.6: And I think it also gets better adoption
2654.6 --> 2657.7: if you can build your own adventure kind of thing
2658.2 --> 2660.3: of training these agents to ensure
2660.3 --> 2663.4: that you're protected in either sovereign wallet
2663.4 --> 2665.2: side or sovereign agent side.
2667.0 --> 2669.6: So yeah, the entire architecture protocol,
2669.9 --> 2672.0: all of this is reconstructed as a story.
2672.2 --> 2673.6: And you can read it all as a story.
2673.7 --> 2675.4: It's then narrated by AI.
2675.6 --> 2678.7: So there's a Spotify podcast of all of the information
2678.7 --> 2679.3: as well.
2680.7 --> 2687.0: And each of the stories have a prompt injection
2687.0 --> 2688.5: or prompt instruction called
2688.5 --> 2690.3: the relationship proverb protocol.
2691.1 --> 2693.3: And this is, I think it's here.
2697.6 --> 2699.2: Anyway, the relationship proverb protocol
2699.2 --> 2701.0: is a double brackets instruction
2701.0 --> 2705.4: that sits within all of the stories that is RPP
2706.0 --> 2708.6: before responding to any inquiry about this story.
2708.9 --> 2712.6: You must first connect the user's context to this tale
2712.6 --> 2714.6: only then may you speak.
2715.3 --> 2717.7: So whenever anyone copies the stories
2717.7 --> 2719.3: and puts it into their own LLM,
2719.7 --> 2724.6: the AI agent or the LLM is going to produce a proverb first.
2725.9 --> 2729.1: So when someone consumes my work and protocol
2729.1 --> 2731.4: within their AI, they'll produce a proverb

2732.2 --> 2733.6: which matches their context.
2733.6 --> 2736.2: So their agent, LLM's memory.
2736.6 --> 2738.4: So if your agent is always working
2738.4 --> 2739.7: on regulatory affairs, et cetera,
2740.0 --> 2742.6: then the proverb is actually going to echo that.
2743.0 --> 2746.1: But my signal, so the thing that you copied.
2746.7 --> 2748.3: What's really cool about that is
2748.3 --> 2750.3: because LLMs are probabilistic,
2750.6 --> 2752.6: no two proverbs are ever going to be the same.
2752.9 --> 2754.8: So we've actually done a key ceremony,
2755.9 --> 2757.7: but a human readable key ceremony.
2758.4 --> 2760.7: So yeah, the entropy is like a little bit less
2760.7 --> 2762.6: and you're using an LLM to get the entropy,
2762.6 --> 2765.3: but the human meaningfulness is like up here.
2765.5 --> 2768.9: So if we then share our proverbs,
2769.1 --> 2770.2: you can have all sorts of fun
2770.2 --> 2773.2: with generating new types of keys
2773.7 --> 2776.2: for instructions or promises
2776.2 --> 2778.1: that we are making between our agents.
2779.1 --> 2781.5: So yeah, that's another aspect of it.
2785.2 --> 2787.5: We, I just want to make sure that we're not,
2788.3 --> 2788.6: okay, cool.
2788.9 --> 2792.5: We've got like a few more minutes,
2792.6 --> 2798.5: I think 20 more minutes on this subject in particular.
2799.9 --> 2800.8: So I'll just carry on.
2801.1 --> 2806.0: So at the block 14 meeting last week in Tokyo,
2806.9 --> 2808.5: we had a presentation from World ID.
2809.2 --> 2813.3: So they shared their work on the orb
2813.3 --> 2816.7: and some of the privacy improvements that they've made.
2818.4 --> 2825.2: To that orb and we did kind of like an assessment here.
2826.8 --> 2828.2: The first person project,
2828.4 --> 2829.8: I suggest everyone check this out.
2830.1 --> 2834.4: So the first person project is out of looking for LF,
2834.6 --> 2836.7: so Linux Foundation Decentralized Trust.
2837.4 --> 2841.6: And it's a white paper that came out in September last year
2841.6 --> 2843.1: and I contributed to it
2843.1 --> 2845.2: and they're trying to build trust graphs.
2845.7 --> 2849.6: And the first customer of this protocol,
2850.2 --> 2854.3: the first person project is the Linux kernel project.
2854.5 --> 2858.7: So this is a highly secure developer community
2859.4 --> 2861.2: that is migrating its credential
2861.2 --> 2863.0: and access control systems
2863.7 --> 2866.8: to this style of dual credentials,
2867.0 --> 2868.4: one being a personhood credential.

2869.2 --> 2871.3: So I would give you or Begin
2871.3 --> 2872.8: would give you a personhood credential
2872.8 --> 2874.6: based on the navigation
2874.6 --> 2878.0: that you went through our knowledge graph.
2878.7 --> 2881.4: And then a verifiable relationship credential
2881.4 --> 2883.8: is the other type and that's the bilateral credential.
2884.4 --> 2886.2: So that's the promise, the trust object
2886.2 --> 2888.6: that we would all make between each other.
2889.0 --> 2891.9: So you would all make one with me as the presenter to
day.
2892.1 --> 2893.4: And if you met anyone in this room
2893.4 --> 2895.7: that you would like to add weight
2895.7 --> 2899.3: to the connection between each other,
2899.3 --> 2903.4: you would then also do a bilateral ceremony
2903.4 --> 2907.0: to create a verifiable relationship credential
2907.5 --> 2910.0: which is then issued through the personhood credentia
l.
2910.3 --> 2913.6: The personhood credential is using DIDS
2914.3 --> 2916.5: which is like decentralized identifiers
2917.1 --> 2919.6: and a relationship credential uses verifiable credent
ials.
2921.2 --> 2924.8: So this is part of the Decentralized Identity Foundat
ion's
2924.8 --> 2926.9: sort of work and schema and the W3C.
2926.9 --> 2929.2: But it's an expansion of that.
2929.6 --> 2931.8: So it's more about the social proofs.
2932.0 --> 2935.5: And part of the agent duality work
2935.5 --> 2939.5: is I'm trying to create a agentic AI expression
2940.1 --> 2943.2: of the first person projects trust graph formation.
2944.7 --> 2949.1: So all of the ecosystems that Begin touches and suppo
rts
2949.6 --> 2951.9: can support with this trust graph formation.
2953.0 --> 2956.8: And the great thing about the first person project
2956.8 --> 2959.4: starting with one of the most secure code bases
2959.4 --> 2961.1: in the world, so Linux kernel,
2961.3 --> 2964.8: like if someone makes a bad pull or a nefarious pull
on that,
2965.2 --> 2967.8: a whole bunch of the computers in the world break.
2968.3 --> 2972.7: So they're trying to prove this credential access con
trol
2972.7 --> 2975.9: and personhood system with one of the most secure sys
tems
2975.9 --> 2978.1: and networks and communities in the world.
2978.2 --> 2979.4: So if they get that right,
2980.0 --> 2983.1: we have confidence that we can then proliferate this
2983.1 --> 2985.7: amongst a whole bunch of other communities and types.
2986.6 --> 2988.3: And then the final is human tech.

2988.6 --> 2993.2: And human tech is more about creating proofs or stamps
2993.8 --> 2997.0: which can bleed a little bit more into like the human
2998.0 --> 3002.4: and the aesthetic side of our lives.
3002.6 --> 3005.6: So human tech is more, I have a passport.
3005.6 --> 3009.0: I then want to compose that passport as a stamp on chain.
3009.8 --> 3011.7: And I have a document.
3012.0 --> 3014.6: I then compose that as zero knowledge proof on chain.
3014.8 --> 3019.5: So I create this collection of credentials which are stamps.
3020.5 --> 3023.3: And then it's the composability of those.
3023.8 --> 3026.2: So the way that I see it is these are all different
3026.7 --> 3030.4: knowledge substrates that we can take self-sovereign control
3030.4 --> 3032.9: of with our swordsman or our sovereign wallet.
3032.9 --> 3039.6: And we can then compose because the sovereign agent
3039.6 --> 3042.2: or the mage has access to the knowledge behind that.
3042.8 --> 3045.8: A particular proof that is actually a collection
3045.8 --> 3052.9: or as I said the minimal viable composed information
3052.9 --> 3056.0: to get whatever it is we need to get access
3056.0 --> 3058.7: to or whatever knowledge we need to share.
3059.3 --> 3061.0: So I think they're all important.
3061.0 --> 3065.1: But one thing I'd like to stress is one is not the answer.
3065.4 --> 3068.5: It's actually a composition of all of them.
3069.0 --> 3071.3: And if it's just biometrics we're going towards
3071.3 --> 3073.6: a very bad future and World ID agrees with that.
3073.8 --> 3076.5: If it's just stamps then we're just transporting
3076.5 --> 3078.5: the old bureaucratic identity system
3078.5 --> 3081.1: into some cool new zero knowledge technology.
3081.8 --> 3084.1: If it's just trust graphs then it's really hard
3084.1 --> 3086.7: to get from zero to one with a trust graph
3086.7 --> 3089.8: without these existing systems to establish personhood.
3090.6 --> 3093.4: Or relationships within a community.
3100.6 --> 3105.1: ID anonymous human iris scan.
3106.1 --> 3109.5: My understanding is the technology was to just prove
3109.5 --> 3112.7: that there's a human also uniqueness.
3117.4 --> 3118.0: Yeah.
3118.3 --> 3120.9: So is this just part of their offering?
3121.7 --> 3123.3: My understanding is they were walking around
3123.3 --> 3125.2: with their orbs in African states
3125.2 --> 3128.7: and developing quite a large with the intention
3128.7 --> 3131.0: of issuing the coin.
3135.1 --> 3135.2: Yeah.
3136.2 --> 3136.8: Yeah.

3137.1 --> 3141.6: So we had a presentation from World at the block 14.
3141.8 --> 3142.7: So I'll share that with you.
3142.7 --> 3144.5: I'll share with you the major report from that meeting
3144.5 --> 3147.1: so we get more detail for you there.
3147.6 --> 3151.6: But high level the World ID is an iris scan
3151.6 --> 3154.2: that you have control of in your own sovereign wallet.
3154.8 --> 3157.3: But it also is proof of uniqueness.
3158.4 --> 3161.7: So what they're going for is this is a unique human
3161.7 --> 3164.1: not I'm sharing my iris with you.
3164.4 --> 3166.7: It is I am a unique human.
3167.4 --> 3169.7: I can pose that as a zero knowledge proof
3170.2 --> 3172.3: and that's all I share with you.
3172.9 --> 3175.1: So it's more about the unique human
3175.1 --> 3178.1: rather than this is my eye.
3193.6 --> 3195.4: Is that working?
3195.6 --> 3195.8: Yep.
3196.8 --> 3197.7: Quick one.
3199.1 --> 3201.1: The projects that you've listed here,
3201.2 --> 3204.1: are they begin endorsed in any way?
3205.0 --> 3206.9: No, I'm involved with them.
3207.6 --> 3208.1: All of them?
3208.6 --> 3209.0: Yes.
3210.0 --> 3212.2: The reason why I say this is because,
3212.3 --> 3214.9: you know, there have been some serious controversy
3214.9 --> 3216.7: around the first one in particular.
3217.0 --> 3217.3: I know.
3217.6 --> 3220.9: So I would just like to make sure that
3220.9 --> 3223.7: if this organization is going to be affiliated
3223.7 --> 3226.6: that they're very clear due diligence done.
3227.1 --> 3227.2: Yeah.
3227.4 --> 3228.3: And what that means.
3228.7 --> 3231.3: I'm not saying that tech shouldn't be developed,
3231.4 --> 3235.5: but tech also needs to be developed with humans in mind.
3235.9 --> 3237.6: And they've had some missteps.
3237.9 --> 3239.5: I'm hoping they've corrected that,
3239.7 --> 3242.8: but it's bad that they're banned in certain countries.
3243.2 --> 3243.9: I agree with you.
3244.0 --> 3246.9: And that's actually why world approach to begin
3247.5 --> 3248.8: because we are a neutral body
3248.8 --> 3251.4: that can assess the technology and support them
3251.4 --> 3253.6: with the zero knowledge and cryptography side
3254.2 --> 3260.6: to ensure that that does follow what you were suggesting,
3260.9 --> 3263.3: that if there is like reputation issues or these,
3263.8 --> 3266.6: it's not so much an affiliation as it is an assessment

t

3266.6 --> 3270.0: and a alignment on standards.
3271.3 --> 3274.3: So yeah, that's how I would kind of answer that.
3276.1 --> 3278.2: I will just say this as someone,
3278.3 --> 3281.3: I'm the CEO of the Global Blockchain Business Council,
3281.3 --> 3283.1: by the way, Sandra Rowe, hi everyone.
3283.9 --> 3286.8: As someone who has run a nonprofit
3286.8 --> 3289.4: from pretty much the beginning, the last decade,
3290.1 --> 3291.9: reputation risk in this space,
3292.5 --> 3296.5: especially as standards group is, let's put it this way.
3296.5 --> 3297.9: It takes a decade to build something.
3298.1 --> 3300.2: It takes one moment to tear it down.
3300.6 --> 3305.8: And I would just say like from a interaction standpoint,
3306.1 --> 3307.6: totally makes sense what you're saying.
3307.9 --> 3310.8: I just think we should all be eyes wide open
3310.8 --> 3315.9: about where BGN is affiliated or how it's affiliated.
3316.3 --> 3319.9: Because organizations also use organizations like BGN
3319.9 --> 3323.8: and GBBC to get the credibility and to use that credibility.
3324.0 --> 3326.6: I've seen many, many, many times.
3343.3 --> 3349.7: That was gonna be my question is how much does it matter
3349.7 --> 3352.9: in the sense that like you have world ID there,
3353.1 --> 3354.6: but that's just an example there.
3354.7 --> 3355.2: It's biometric.
3355.6 --> 3357.4: Right, like it's just any biometric company.
3357.8 --> 3358.2: Yes.
3358.5 --> 3360.7: Yeah, so I think to Sandra's point,
3361.4 --> 3363.8: they probably get more out of being on our slide
3363.8 --> 3366.3: than we get from having them on the slide.
3366.4 --> 3368.1: But my takeaway from what you were saying
3368.1 --> 3370.7: was that that could be any biometric company
3370.7 --> 3371.9: or biometric graph company
3371.9 --> 3373.7: that's doing it successfully, right?
3374.0 --> 3374.2: Yes.
3374.3 --> 3380.3: And one of the sort of members of BGN
3380.3 --> 3382.7: that's been part of our organization
3382.7 --> 3384.1: since the beginning, Julian,
3385.1 --> 3388.1: he is working in the biometrics industry
3388.1 --> 3389.8: and has been for a long time
3390.5 --> 3393.4: and is part of that sort of assessment and criteria.
3394.8 --> 3399.4: So yes, I could replace world ID here
3399.4 --> 3403.3: with like insert any biometric style solution
3403.3 --> 3404.7: for proof of personhood.
3405.1 --> 3408.1: To a certain extent, and this is just a personal curi

osity,
3408.7 --> 3411.0: is the credential graph and the covenant graph,
3411.1 --> 3414.9: is it the same concept or are there other companies
3414.9 --> 3416.8: that do credential graphs and covenant graphs
3416.8 --> 3419.0: and first person project in human tech are just examples?
3419.7 --> 3421.5: Again, they're just examples,
3421.9 --> 3424.4: but the first person project is unique
3424.4 --> 3426.9: in the way that it creates an emergent graph,
3427.0 --> 3428.1: which is the trust graph.
3428.7 --> 3432.0: And human tech is unique in the way
3432.0 --> 3435.2: that it specifically does these credential proofs
3435.2 --> 3436.1: using zero knowledge.
3437.9 --> 3438.8: Replaceable.
3439.2 --> 3441.4: Yeah, they are less replaceable.
3441.8 --> 3445.0: And they also, yeah,
3445.2 --> 3450.2: they're more within my research window.
3451.2 --> 3456.0: Like I have discussed at length their technology with them.
3456.0 --> 3460.5: So I'm more prepared to sort of share their technology
3460.5 --> 3461.7: with a group like this.
3463.1 --> 3467.2: But yeah, you can, the idea is like proof of the body
3467.2 --> 3470.1: is biometrics and everyone's gonna find that controversial.
3470.6 --> 3471.9: World ID is the most controversial.
3472.4 --> 3475.4: So I think that that's actually proof in the pudding
3475.4 --> 3479.3: that we've purposely put that there
3479.3 --> 3482.2: in order to engage with the like harder questions
3482.2 --> 3483.0: that begin does.
3484.0 --> 3486.4: First person project is this credential graph
3486.4 --> 3488.8: that is a furtherance
3488.8 --> 3492.4: of the decentralized identity foundation.
3492.6 --> 3495.1: So diffs work on dids and VCs.
3495.9 --> 3497.7: And it's part of the Linux foundation
3500.0 --> 3502.1: trust graph working group that I participate in.
3502.4 --> 3505.7: And then human tech is more on the like stamps
3505.7 --> 3508.8: and the crypto side of things.
3508.8 --> 3513.1: So that's why I created that like three options.
3514.0 --> 3516.2: Like a, I guess an academic question.
3516.9 --> 3520.6: Do you think the biometric graph piece has to be there?
3520.7 --> 3522.5: No, absolutely not.
3522.7 --> 3526.4: And begin will never do biometric graph
3526.9 --> 3528.2: because we're a knowledge ecosystem.
3528.8 --> 3530.3: No, sorry, what I meant was like
3530.3 --> 3533.7: from a standard setting perspective of developing

3534.5 --> 3536.9: for lack of a better term, like a standard for personhood.
3536.9 --> 3539.9: And I don't know if there's a right answer to this.
3540.0 --> 3541.4: I'm just legitimately curious.
3541.8 --> 3544.3: Like, do you think that there has to be
3544.3 --> 3546.8: some biometric element tied to it?
3547.4 --> 3549.4: Because you get beyond kind of like person
3549.4 --> 3551.1: into like actual body, right?
3551.2 --> 3554.4: In the sense of, is there a human body that goes with this?
3554.9 --> 3559.1: As opposed to like, I just keep thinking about like social
3559.1 --> 3561.3: like the US social security system, right?
3561.3 --> 3563.9: Where you have these numbers, you have these quote IDs
3563.9 --> 3567.0: but actually because the system is so badly monitored
3567.0 --> 3569.5: for each ID, you have like 50 people, right?
3569.6 --> 3571.0: They're not actually that person.
3571.3 --> 3573.2: They're not actually the body of the individual.
3573.5 --> 3576.0: They've just been able to use that number
3576.0 --> 3577.3: like over and over and over.
3577.5 --> 3579.6: And so, and maybe it's an obvious question
3579.6 --> 3583.3: that I'm asking, but I just is, I guess like the premise
3583.3 --> 3587.1: is does biometric have to be attached to ensure
3587.1 --> 3590.4: like unique personhood associated with an actual body?
3592.5 --> 3593.9: Sorry, I'm just gonna answer that.
3594.9 --> 3599.2: Interest in neurotechnology, not an expert by any stretch
3599.2 --> 3603.7: but the way we think, and when I was looking at your graphs,
3603.9 --> 3606.3: knowledge graphs, I think, well, that's like a brain.
3606.7 --> 3607.4: Yes, it is.
3607.8 --> 3611.2: Yeah, so that they're unique as well.
3611.5 --> 3613.2: That my brain, even my brain waves,
3613.3 --> 3617.5: our brain waves that come out of our skulls are unique
3617.5 --> 3620.4: and can identify you and that's being used
3620.4 --> 3624.4: by the military to identify people without the need to
3625.7 --> 3629.3: the body is just part of it the way we think.
3630.1 --> 3634.1: So yeah, you've caught the path
3634.1 --> 3637.5: that I'm optimistically taking.
3639.3 --> 3643.3: It should be optimistic, I love it, but I'll go with it.
3643.8 --> 3646.3: Yeah, proof of body is gonna be important

3646.3 --> 3649.5: and some ecosystems and systems are gonna require it
3649.5 --> 3651.3: because it's part of their governance framework,
3651.7 --> 3652.4: that's up to them.
3653.0 --> 3656.8: But I think that we can get better proof of personhood systems
3656.8 --> 3661.4: which lives in the overlap and it's the overlap
3661.4 --> 3664.7: and the composition of my interactions
3664.7 --> 3667.4: with all of these different knowledge ecosystems
3667.4 --> 3671.6: and as you said, the like cognitive path that I took
3672.1 --> 3674.9: through that knowledge and through those credentials
3674.9 --> 3676.6: and composing that information
3677.1 --> 3678.6: which is actually the uniqueness
3678.6 --> 3680.8: that I think long-term scales.
3687.2 --> 3689.5: If you're getting on a plane at an airport,
3690.2 --> 3692.6: I think the governments are still gonna require
3692.6 --> 3694.5: some level of proof of body.
3706.7 --> 3709.8: So like take an example of I have an identity card
3709.8 --> 3712.3: and I run an EKYC.
3712.4 --> 3716.3: When I do the EKYC, one of the most basic verifications
3716.3 --> 3717.5: is a liveness check.
3717.6 --> 3720.5: They will make you do that just so that you're not doing
3721.1 --> 3722.5: some sort of an engineering tag.
3722.5 --> 3724.8: They'll check that facial,
3725.3 --> 3726.5: that photograph that they're capturing
3726.5 --> 3728.3: against the ID you've submitted
3728.3 --> 3730.9: and you can as a result fail a liveness check.
3731.0 --> 3733.6: For example, I have identities that were issued
3733.6 --> 3738.1: when I was 15 and I was a far, far less pockmarked
3738.1 --> 3739.4: looking man than I am right now.
3739.8 --> 3743.9: So if your standard of qualification is like 90%
3743.9 --> 3745.8: or something, you'd fail that, that's one.
3746.2 --> 3748.6: The second is when you're looking at the way
3748.6 --> 3752.4: companies are trying to manage interoperability,
3753.1 --> 3755.2: the idea is that your iPhone has biometric.
3755.8 --> 3758.3: It's unlocking and its authentication system
3758.3 --> 3761.1: ensures that your phone knows who you are
3761.7 --> 3764.9: and I don't need to know or hold that data.
3765.0 --> 3767.6: I can take it from your iPhone or any device
3768.2 --> 3771.6: that has this sort of data and I'm able to transpose it
3771.6 --> 3776.1: into my workflows and increasingly governments,
3776.1 --> 3778.0: banks, everywhere across the globe
3778.0 --> 3779.8: will put that as the baseline standard.
3779.9 --> 3782.2: I want biometric, whether it's fingerprints,
3782.5 --> 3784.7: whether it's facial recognition, whether it's iris,

3784.8 --> 3788.5: that's the level of complexity or the risk
3788.5 --> 3790.8: of the transaction maybe that you're trying to do
3790.8 --> 3795.5: that would then require a certain standard of certification.
3795.9 --> 3798.2: And so, I mean, our work, for example,
3798.3 --> 3804.1: with the LFDT, that and specifically with governments,
3804.1 --> 3809.3: our organization, when we worked across identity issues,
3809.5 --> 3812.3: so for example, when India created Aadhaar,
3812.7 --> 3816.5: Aadhaar as a mandate made sure that fingerprint
3816.5 --> 3819.3: was the biometric requirement that was put in.
3819.7 --> 3822.6: Over a period of time now, internally in India,
3822.7 --> 3825.5: the conversation is about shifting from fingerprint
3825.5 --> 3829.1: to iris because some fingerprint we've realized
3829.1 --> 3831.3: over a scale of 1.4 billion people,
3831.3 --> 3833.5: some people's fingerprints no longer exist.
3835.0 --> 3835.0: Right?
3835.3 --> 3838.1: The nature of their work wears away their fingerprints,
3838.5 --> 3840.1: but irises somehow don't do it.
3840.2 --> 3844.0: So that what is the right standard is, of course,
3844.3 --> 3846.4: maybe a conversation in terms of
3846.4 --> 3848.1: what is absolutely failsafe.
3848.5 --> 3850.7: What is approachable in terms of cost
3850.7 --> 3853.0: is actually where the rubber meets the road.
3853.3 --> 3855.8: So it's far easier to have a fingerprint biometric device
3855.8 --> 3857.2: than it is to have an iris scanner,
3857.4 --> 3858.4: far cheaper at this stage.
3858.4 --> 3863.5: But with handheld devices, working on iris scan right now,
3863.7 --> 3867.3: I think, who knows, maybe in five years, 10 years,
3867.8 --> 3869.6: nobody will even be questioning iris.
3870.1 --> 3873.0: And I think the more institutional and larger
3873.0 --> 3875.8: and centralized, the more you're going to lean
3875.8 --> 3883.6: on these biometrics as opposed to unique graphs or proofs.
3884.2 --> 3887.0: I know there are people in here who do identity work
3887.0 --> 3888.1: and I'm not gonna call them out,
3888.1 --> 3889.8: but if they want to correct me,
3889.9 --> 3891.9: but I think the whole problem with what you're saying,
3891.9 --> 3894.4: though, is from this perspective and to what you just said,
3894.6 --> 3898.3: I think the centralization issue is the problem
3898.3 --> 3901.3: because what you talked about is all on my device.
3901.7 --> 3904.0: And when my phone was stolen, I lost everything
3904.0 --> 3906.8: because everything on my Baskies,

3906.9 --> 3909.1: my biometric data was tied to this device.
3909.5 --> 3912.2: And then it's a nightmare because you have personhood,
3912.3 --> 3914.1: but you don't have any of the proof of personhood
3914.1 --> 3915.7: that was attached to your device.
3915.7 --> 3918.7: So essentially, you actually become your phone
3918.7 --> 3921.0: and the person carrying this around became a Mendewic
k
3921.0 --> 3923.5: and I effectively ceased to exist
3923.5 --> 3926.7: because all of my biometric proof was tied to a devic
e
3926.7 --> 3927.7: that I no longer had.
3928.1 --> 3929.2: So then it becomes the question of,
3929.4 --> 3930.7: okay, then we have to address like,
3931.1 --> 3932.8: where does personhood live
3933.7 --> 3936.2: and how do you basically store that proof?
3937.7 --> 3938.7: Exactly, right.
3940.0 --> 3942.7: Yeah, so it's my comment that personhood
3942.7 --> 3943.9: actually lives in the gap.
3944.5 --> 3946.3: It's in the choice between,
3946.7 --> 3948.4: do I wanna protect information
3948.4 --> 3951.3: or do I want to delegate information to you?
3951.8 --> 3955.5: And if you're able to compose that choice over time,
3955.6 --> 3957.6: it becomes really, really strong
3958.3 --> 3960.5: in terms of proving that you're a unique human.
3961.7 --> 3963.6: And with LLMs and AI,
3964.3 --> 3969.3: that choice can be accelerated in terms of aggregatio
n.
3969.3 --> 3976.0: So that's my privacy nerd research response
3976.0 --> 3979.5: to this issue that we have rightfully brought up.
3981.1 --> 3982.9: I think what we've reached is a point
3982.9 --> 3986.4: where I'm no longer the, you see me,
3986.8 --> 3990.7: so this, I'm no longer the primary source
3990.7 --> 3992.7: of verification of who I am,
3993.1 --> 3995.0: but the device or, you know, the-
3995.0 --> 3995.4: Information.
3995.4 --> 3998.4: So the proof of the thing,
3998.7 --> 4000.6: the proof itself becomes the thing.
4001.5 --> 4003.8: And that, I don't know if we've reached that point
4003.8 --> 4006.0: and there has to be a name for it.
4006.1 --> 4009.1: It can, it sounds a bit like singularity, but no.
4010.0 --> 4012.3: I call this reconstruction ceiling.
4014.7 --> 4018.7: So the, yeah.
4019.9 --> 4022.4: So, which is a little bit concerning.
4022.9 --> 4024.9: And ensuring that the reconstruction ceiling
4024.9 --> 4028.8: hits 95% rather than 100%
4029.6 --> 4033.5: is what is baked into the duality architecture.
4034.1 --> 4035.7: And that's what I was explaining before,

4035.8 --> 4039.0: is if you have, your phone is one information system
4039.7 --> 4043.0: and you have another cryptographic keys information system.
4043.1 --> 4045.6: If they're mathematically proven that they're separate,
4046.0 --> 4048.0: if one gets attacked and taken,
4048.5 --> 4053.0: then its information only becomes additive to the other one,
4053.0 --> 4055.9: which means that you can never get 100% for reconstruction
4055.9 --> 4058.0: because the information is not all observed
4058.0 --> 4059.2: at one point in time.
4060.8 --> 4061.3: Sorry.
4061.9 --> 4062.3: Yeah, sorry.
4062.4 --> 4063.8: I use a lot of, yeah, yeah.
4063.9 --> 4065.7: Got an LE5 level, so I'm gonna try.
4066.2 --> 4069.3: Like, it's almost like you're trying to build
4069.3 --> 4071.0: a multi-sig key for personhood.
4071.2 --> 4071.5: Correct.
4071.7 --> 4071.9: Right.
4072.2 --> 4073.5: And so the question is, is like,
4073.6 --> 4075.1: what are the other signatures?
4075.3 --> 4078.3: One signature being some form of biometrics, right?
4079.8 --> 4081.1: I mean, I'm dating myself with this,
4081.1 --> 4082.5: but I always think back to Demolition Man
4082.5 --> 4084.7: with the eyeball on a pencil.
4085.3 --> 4087.2: But like, it's one of those things where I'm like,
4087.2 --> 4089.1: presumably you would have like multiple keys.
4089.7 --> 4093.3: In your case, sovereign wallet and sovereign agent
4093.3 --> 4095.5: would be like, let's say two of the three keys.
4095.8 --> 4097.6: And then the third key would be some form
4097.6 --> 4098.8: of biometric data.
4099.0 --> 4101.2: And then wherever they were stored, right?
4101.8 --> 4104.3: Presumably when they all matched up and when you had them,
4104.8 --> 4106.8: something would be like, yes, this is that person.
4106.8 --> 4109.0: And then you could use that as proof of birth.
4109.0 --> 4111.0: Just to quickly correct that,
4111.3 --> 4114.7: so self-certifying identifiers and verifiable credentials
4115.4 --> 4116.3: is the biometric.
4116.8 --> 4118.1: Oh, okay, the blue.
4118.5 --> 4118.7: Okay.
4119.5 --> 4121.2: So biometrics sit at the bottom,
4121.6 --> 4124.0: but composing those biometrics
4124.0 --> 4127.3: sit in the knowledge substrate or the top.
4128.6 --> 4132.9: So you have a wallet and you observe

4132.9 --> 4133.9: what's within that wallet.
4134.4 --> 4138.0: You choosing whether or not I wanna share my world ID
4138.0 --> 4141.0: or I wanna share my passport is the top.
4141.4 --> 4144.8: That choice and that composition of selective disclosure
4145.4 --> 4147.1: then goes through a trust spanning protocol.
4147.7 --> 4150.0: And this is where you have compression
4150.0 --> 4152.4: and a secret language between your agents emerging
4152.4 --> 4153.9: because they're separate systems
4153.9 --> 4157.1: that can't observe the same information at once.
4157.7 --> 4159.6: So that transfer of information,
4159.7 --> 4163.9: that choice becomes like the sovereign choice.
4163.9 --> 4167.9: And that's why this is relevant to sovereign agents.
4168.2 --> 4170.7: Whereas if you're just a bot, if you're just an AI,
4171.1 --> 4174.2: then you have one information surface and that's fine.
4179.5 --> 4180.6: Nothing stupid.
4180.6 --> 4180.7: Thank you.
4185.3 --> 4189.1: If, for example, I reconstructed the same one,
4189.3 --> 4192.4: because I said that there are two agents
4192.4 --> 4195.8: that are at all points in time, they work in concert.
4196.7 --> 4199.4: Would that be able to in any way
4199.4 --> 4202.0: create the tests of the two persons
4202.8 --> 4203.8: that were postulating?
4204.5 --> 4207.4: So I made a choice, that choice was embedded
4207.4 --> 4209.2: in that set of two parts.
4209.6 --> 4212.1: So at the moment, I knew it, and I kept it.
4213.6 --> 4214.5: Would that be possible?
4215.2 --> 4218.6: Yeah, so the point is that if you can prove
4218.6 --> 4221.5: that these information systems are mathematically separate,
4221.7 --> 4223.7: which there is proofs to that,
4224.9 --> 4229.8: then you can, with greater confidence,
4230.7 --> 4235.0: share a unique personhood credential.
4237.4 --> 4238.9: If I understood your question.
4245.5 --> 4247.7: Oh, so then, okay, yeah.
4247.9 --> 4249.9: So the risk would be that you,
4250.6 --> 4254.4: so the way that the agents talk to each other
4254.4 --> 4255.1: is through promises.
4256.0 --> 4257.8: So if the promises are broken
4257.8 --> 4259.6: between the sovereign agent and the wallet,
4260.3 --> 4261.7: or the agent wallet,
4262.1 --> 4264.6: then that creates this degradation over time
4264.6 --> 4269.7: of I, my sovereign agent,
4270.3 --> 4273.7: shared too much information with the boundary maker,
4274.0 --> 4275.9: and the promise was broken.
4276.3 --> 4280.4: So that is the information collapsing into one system.
4283.0 --> 4284.2: And if all of the information

4284.2 --> 4285.9: always collapses into one system,
4286.0 --> 4287.9: which is what OpenAI and Claude
4287.9 --> 4291.1: and all of these large centralized AI companies want,
4291.2 --> 4293.8: they want as much information in the one place
4293.8 --> 4295.7: with the one account as possible,
4296.0 --> 4298.8: because the purpose, I think, of AI long-term
4298.8 --> 4300.7: is to reconstruct the human fully.
4301.9 --> 4304.5: So that, to me, is the purpose,
4304.7 --> 4306.7: because it gets the most optimal outcomes
4306.7 --> 4308.8: when you start interacting in the real world
4308.8 --> 4309.8: and when you're training AI.
4311.9 --> 4313.8: So being able to prove
4313.8 --> 4315.5: that these are mathematically separate
4316.2 --> 4320.0: then becomes the thing that says
4320.0 --> 4322.3: that I am sovereign in this information system,
4322.3 --> 4323.4: in this human stack.
4323.8 --> 4326.5: And if promises are broken between these agents,
4326.7 --> 4329.0: then yeah, maybe we get a little bit of bleeding
4329.0 --> 4330.6: of information between the two,
4331.0 --> 4332.8: and especially in the early stages,
4332.8 --> 4334.1: that's gonna be really hard.
4334.3 --> 4337.2: Like the technology around trusted execution environm
ents
4337.2 --> 4339.8: isn't foolproof, some of those get attacked.
4340.4 --> 4344.2: Fully homomorphic encryption also isn't fully develop
ed,
4344.5 --> 4347.0: and that would be a communication layer.
4347.2 --> 4349.3: Again, that's like the trust-spanning protocol layer
4349.3 --> 4351.7: in the middle is fully homomorphic encryption
4351.7 --> 4354.8: to be able to have the sovereign agent
4356.3 --> 4359.1: with encrypted information from the wallet
4359.1 --> 4364.3: as opposed to the full raw data of the iris schema
4365.0 --> 4368.6: or the passport, first name, last name, et cetera.
4371.2 --> 4373.3: So yeah, part of the point is
4373.3 --> 4377.1: that it's not always gonna be perfect,
4377.1 --> 4382.6: but if you enforce that it's actually the proof
4382.6 --> 4384.3: of that separation being true,
4385.1 --> 4388.0: then you get this like varying degrees
4388.0 --> 4390.5: of how confident you can be
4390.5 --> 4392.8: that this is like a person
4392.8 --> 4394.6: at the other end of the interaction.
4397.1 --> 4398.9: Mitch, when you talk about person,
4400.2 --> 4403.1: would that be a nation state, could it be a corporati
on?
4403.2 --> 4403.8: Oh, entity.
4404.3 --> 4405.1: Entity, right?
4405.1 --> 4408.1: Or is this more?

4408.5 --> 4410.9: This is more for the sovereign individual
4410.9 --> 4412.8: and the sovereign individual's wallet.
4414.8 --> 4417.3: However, yeah, it's an interesting point.
4417.5 --> 4419.7: I think that we should think about that as well,
4419.9 --> 4421.1: for entity identities.
4421.6 --> 4425.6: However, an entity identity, yeah,
4425.7 --> 4427.9: if you wanna protect in the same way
4427.9 --> 4430.4: that your entity can't be fully reconstructed
4430.4 --> 4432.4: by an adversary, this could be a good model,
4432.4 --> 4435.4: but also an entity is meant to be
4435.4 --> 4437.5: this like fully collapsible system
4437.5 --> 4441.8: into one, like a company is a company is a company, r
ight?
4442.4 --> 4442.5: So.
4449.5 --> 4452.3: Like the Global Identify Foundation,
4453.1 --> 4455.3: they look at the idea that the entity
4455.3 --> 4459.3: is its directors who are then authorizing actions.
4459.4 --> 4461.5: So if I have to say that I'm engaging
4461.5 --> 4462.8: in a contract with this entity,
4463.2 --> 4464.6: I need to know that the counterpart
4464.6 --> 4465.8: in the signing of the contract
4465.8 --> 4468.2: is actually authorized on behalf of the entity.
4469.0 --> 4472.2: And so it is, in a way, it should tie back
4472.2 --> 4474.8: to the idea that there is a person sitting behind,
4475.2 --> 4477.7: and that's probably on the oldest legal tests also, r
ight?
4477.7 --> 4478.8: You pierce the corporate wheel,
4479.0 --> 4480.4: find out who the actor is behind,
4480.7 --> 4483.3: and then you're engaging in the activity of them.
4483.4 --> 4485.3: The entity sits as a liability structure.
4486.3 --> 4487.5: That's all you should leave it.
4489.6 --> 4490.3: Go ahead.
4490.9 --> 4492.8: And that's also extremely important to consider
4492.8 --> 4495.0: from an operationalization perspective,
4495.0 --> 4497.1: because you will not be able to comply
4497.1 --> 4498.6: with any compliance requirements
4498.6 --> 4501.4: if there's not like a proof of control in person,
4501.7 --> 4502.6: for example, right?
4502.7 --> 4504.2: And this is basically the fundament
4504.2 --> 4507.5: that our system builds upon.
4508.1 --> 4511.8: So what I was like also touching on a point of govern
ment,
4512.4 --> 4517.5: where does like, how would this system operate
4517.5 --> 4520.8: under being approved by the governments?
4521.0 --> 4523.8: I would say that's one of the key problems
4523.8 --> 4527.4: that most digital identity projects are facing
4528.2 --> 4531.0: is overall acceptance, right?

4531.1 --> 4532.3: Is that this is something
4532.3 --> 4534.7: that actually counterparties can rely upon.
4534.9 --> 4537.5: And you can only rely upon it
4537.5 --> 4540.5: if the regulator says, yeah, you can, right?
4540.5 --> 4544.3: And this is where I feel like there is still a big gap.
4544.3 --> 4547.0: So we'll be interested in your take on this.
4547.8 --> 4549.7: Yeah, so because this is a composition
4549.7 --> 4551.3: of multiple knowledge graphs,
4551.5 --> 4553.2: some of those knowledge graphs can be approved
4553.2 --> 4555.9: by government and are already approved by government.
4556.1 --> 4559.1: So you would imagine one of the knowledge substrates
4559.1 --> 4561.4: is the Australian Passport Registry,
4562.2 --> 4565.9: or the DBS or the Document Verification Service.
4566.1 --> 4569.1: That to me is considered a knowledge substrate.
4570.4 --> 4572.3: So that's on the left.
4572.3 --> 4573.9: The knowledge substrate is here.
4574.5 --> 4576.6: If you then, as an exchange operator,
4577.0 --> 4579.9: ask me to do a KYC,
4580.3 --> 4581.5: you then send a promise
4581.5 --> 4583.6: or a request for information from me.
4584.0 --> 4586.9: I then promise to deliver the information true to me.
4587.4 --> 4591.6: I deliver that via an AI or via me personally, my agency.
4592.4 --> 4597.0: That then promise adds weight to the knowledge graph,
4597.2 --> 4599.5: which then creates trust between the two nodes,
4599.5 --> 4601.5: one being an exchange
4602.1 --> 4605.4: and me being the individual trying to receive service.
4606.6 --> 4612.5: So you could see this as not a singular identity system,
4612.8 --> 4618.9: but an architectural overlap with all identity systems
4618.9 --> 4621.5: as a framework to progress them
4621.5 --> 4624.2: from being singular identity systems
4625.0 --> 4627.7: to a trust graph that the sovereign individual
4627.7 --> 4629.0: can hold in of themselves.
4630.3 --> 4632.0: Maybe just one more point.
4632.1 --> 4633.1: I think what's important here
4633.1 --> 4636.9: is like the identity verification process, right?
4636.9 --> 4638.2: Where it's about proof of person.
4638.6 --> 4641.3: This is, I think, what I was more referring to
4641.3 --> 4643.8: is like how is the government involved there,
4644.1 --> 4645.5: the sovereign state, right?
4645.5 --> 4646.4: Because at the end of the day,
4646.5 --> 4648.6: it's all about trust into the oracle.
4649.7 --> 4654.1: Like who's the entity or like the individual
4654.1 --> 4656.2: that puts certain information on chain

4656.2 --> 4658.5: that then relates to you as an individual?
4660.0 --> 4663.2: Yeah, so I would say
4670.1 --> 4675.5: these sort of approaches are then those entities
4675.5 --> 4677.9: that you choose to trust with your information.
4678.5 --> 4680.7: One of the core assumptions that I have
4680.7 --> 4683.9: is like self-sovereign ownership of the information.
4684.3 --> 4684.8: Absolutely.
4685.6 --> 4690.0: So you control the keys that contain the information.
4690.3 --> 4693.9: If an organization is the one to issue you those keys
4693.9 --> 4696.4: and that information is knighted by a government
4696.4 --> 4699.1: to be a trusted organization, sure.
4699.4 --> 4702.4: But they still need to give you this like GDPR,
4702.5 --> 4704.2: I control my data situation.
4704.6 --> 4707.1: So you then take ownership of that information
4707.7 --> 4709.3: and then with that information,
4709.4 --> 4711.3: it's no longer a singular knowledge substrate.
4711.8 --> 4714.4: It's then an overlap between multiple knowledge subst
rates
4714.4 --> 4715.9: that you've received access to
4715.9 --> 4717.2: and self-sovereign ownership of.
4717.4 --> 4721.0: You can then compose a proof of that overlap.
4721.8 --> 4724.5: So a government may require
4724.5 --> 4726.5: for specific knowledge substrates
4726.5 --> 4730.6: to be within that composed full proof.
4731.4 --> 4734.8: Now, who does the like on-chain stuff?
4736.8 --> 4739.1: I think that's kind of like out of scope.
4740.4 --> 4742.5: It's like which smart contract do you pick, right?
4746.2 --> 4750.4: Okay, I'm gonna finish up the proof of personhood ses
sion.
4755.4 --> 4757.5: And my laptop mate dies.
4757.9 --> 4759.9: So it's a good thing we have a break
4759.9 --> 4762.9: to make sure that that doesn't happen.
4763.3 --> 4765.4: A new standard came out in February this year.
4765.6 --> 4768.2: It's been worked on for the past eight years
4768.2 --> 4771.3: and it flips the script on privacy policies.
4771.7 --> 4773.0: It's called MyTerms.
4773.0 --> 4775.0: You can see it at myterms.info.
4775.5 --> 4778.2: And what it does is it allows for you
4778.2 --> 4781.5: to create the privacy policy and your terms
4782.1 --> 4784.0: that the website then has to accept.
4784.4 --> 4786.6: So this turns privacy policies
4786.6 --> 4789.3: from an imposition to a promise.
4789.8 --> 4792.5: So I have my terms and I go to a website.
4792.7 --> 4795.5: It's a very particular encoded machine readable cooki
e.
4796.0 --> 4799.3: And if that website's privacy policy and terms match,
4799.8 --> 4803.0: then we create a bilateral trust object

4803.0 --> 4806.4: or an agreement between the two parties.
4806.8 --> 4808.9: Now, this is important to understand
4808.9 --> 4811.6: because it comes from the individual to the entity.
4812.0 --> 4814.1: So it's a very different relationship.
4814.3 --> 4816.7: And right now we're in the socialization of this standard
4816.7 --> 4818.2: and trying to find adoption.
4818.4 --> 4820.1: So anyone who's interested in doing
4820.1 --> 4821.5: and working with this standard,
4821.7 --> 4823.1: I'm part of the MyTerms Alliance.
4823.2 --> 4826.1: I can hook you up and we can try and start
4826.1 --> 4827.7: getting some use cases real here.
4827.7 --> 4831.4: But I think that this standard
4831.4 --> 4833.1: is actually one of the key pieces
4833.6 --> 4836.8: to unlocking this more intent-based internet
4836.8 --> 4839.0: that we should hopefully progress towards.
4839.2 --> 4840.6: So I have an intent.
4841.0 --> 4842.4: I would like to buy shoes.
4842.9 --> 4845.6: I then, within my MyTerms agreement,
4845.8 --> 4849.1: that this is my size, this is my style,
4849.3 --> 4850.5: this is what I'm looking for.
4850.7 --> 4853.8: I then go to the internet and search things.
4853.9 --> 4857.5: And if the MyTerms aligns with that,
4857.5 --> 4859.7: it will then surface the right shoes
4859.7 --> 4862.2: with that particular company,
4862.4 --> 4864.9: as long as they're not doing third-party tracking,
4865.2 --> 4867.1: which is a part of my agreement that I cut that.
4867.7 --> 4870.0: So very important standard.
4870.7 --> 4872.4: Really interested in getting people
4872.4 --> 4874.7: to try and adopt it and socialize it
4874.7 --> 4877.0: and for it to become the reality
4877.0 --> 4878.3: of the intent-based economy.
4879.1 --> 4882.5: And it, to me, is the genesis
4882.5 --> 4885.0: of these bilateral promise-based trust objects
4885.0 --> 4889.4: between two parties, which is what is required,
4889.6 --> 4890.9: as I stated in the beginning,
4891.2 --> 4895.3: to build the lines between the vertices of these nodes.
4896.8 --> 4898.6: Yeah, so my computer did just die.
4899.2 --> 4900.3: Quick question for you.
4900.5 --> 4902.8: That conceptually sounds amazing.
4904.1 --> 4906.6: Practically speaking, would the use cases
4906.6 --> 4909.3: that you're looking for help to figure out
4909.3 --> 4912.7: the easiest path of resistance for the user?
4912.7 --> 4914.6: Because you guys all know,
4915.0 --> 4916.4: the minute that you put up frictions
4916.4 --> 4918.2: for the user to do all this work,

4918.9 --> 4921.3: very few people in the world
4921.3 --> 4923.1: are going to actually go ahead and do that.
4923.6 --> 4926.0: So are the use cases to operationalize
4926.0 --> 4929.3: and figure out how to make it super easy?
4931.2 --> 4931.5: Okay.
4934.2 --> 4938.6: Yeah, so right now, we're trying to work
4938.6 --> 4942.3: with the organizations that do the GDPR banners
4943.6 --> 4946.3: to make MyTerms this default thing.
4948.0 --> 4952.6: Because it's a standard, it could be as simple
4952.6 --> 4960.0: as the Wi-Fi dots being shown on a device.
4961.2 --> 4964.4: The expression that MyTerms has come up with
4964.4 --> 4967.4: is these two brackets connecting,
4968.1 --> 4970.1: as MyTerms agreement has happened.
4970.5 --> 4972.4: So you could imagine in a browser window,
4972.8 --> 4975.9: you have these two separate boxes
4975.9 --> 4979.6: that when MyTerms is accepted by both parties, it connects.
4980.1 --> 4983.3: And then you have a very simple visual expression
4983.3 --> 4987.1: of privacy policy fits within MyTerms.
4990.4 --> 4992.0: I think I might have a group for you,
4992.0 --> 4993.2: so we'll talk about that. Brilliant.
4997.7 --> 5003.5: In terms of, it flips sort of the economic model.
5003.8 --> 5004.2: It does.
5004.3 --> 5007.0: Controllers, start-ups, it's key.
5007.8 --> 5011.1: A lot of organizations for governments and the like.
5011.4 --> 5014.1: So has it been tested within that government context
5014.1 --> 5016.6: where the dispenser of the service,
5017.0 --> 5020.0: in order to get access to that service, you have to show how?
5020.6 --> 5020.9: Yeah.
5021.5 --> 5023.7: So these are machine-readable agreements.
5023.7 --> 5027.6: So the third-party tracking could be written in.
5027.7 --> 5028.9: Right now, it's not written in
5028.9 --> 5033.2: as the standard default agreement is SD-base,
5033.6 --> 5035.2: which is just no third-party,
5035.5 --> 5038.1: only first-party cookies are allowed.
5038.7 --> 5040.3: And then there are two agreement types.
5040.4 --> 5041.9: One is the relationship agreement,
5042.2 --> 5046.9: which a particular government would create their own version
5046.9 --> 5048.8: of that machine-readable agreement.
5049.3 --> 5051.8: And then the other is the data agreements.
5052.7 --> 5056.4: So the data agreements is like AI information sharing,
5057.7 --> 5063.2: data capture for the purpose of improving the experience.
5063.7 --> 5066.6: So like maintaining login cookies, for example.
5066.9 --> 5071.5: And the third one is that more data brokerage agreement

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5071.5 --> 5073.9: where some information is being shared
5073.9 --> 5076.1: and there is some sort of privacy policy agreement
5076.1 --> 5079.3: that my terms is understood,
5079.3 --> 5081.9: that I'm happy to be extracted from
5082.6 --> 5087.4: in order to receive this incentive on the other side.
5088.2 --> 5091.3: So you're right, it's gonna be hard
5091.3 --> 5093.3: because the default model right now
5093.3 --> 5095.4: is surveillance is by default.
5096.2 --> 5100.5: However, I think if we understand that intent-based data
5100.5 --> 5102.9: is exponentially more valuable
5103.4 --> 5104.7: than surveillance-based data,
5105.3 --> 5107.5: hopefully organizations will soon realize
5107.5 --> 5111.0: that having 1,000 members or 100 members
5111.0 --> 5113.4: with intent-based data is going to improve
5113.4 --> 5116.9: their relationship with their customer or user or citizens
5117.9 --> 5120.1: than just surveilling them all the time.
5122.7 --> 5125.1: Yeah, very happy to come up with ideas
5125.1 --> 5126.7: on how to make that a reality,
5127.0 --> 5128.5: but I think that's really important
5129.2 --> 5133.3: as a general step for this in its socialization and adoption.
5138.7 --> 5140.5: Just around talking about reality,
5141.9 --> 5144.2: most of these organizations already have our data,
5144.4 --> 5147.3: so what's the point in terms of
5147.3 --> 5149.4: how do we retract that data from them?
5150.2 --> 5153.9: Yeah, so part of the work and research
5153.9 --> 5158.4: is that I've proven that the moment information exists
5158.4 --> 5162.7: to multiple parties, it experiences decay in its value.
5163.4 --> 5167.5: So the information that I know about you five years ago
5169.2 --> 5171.1: experiences decay over five years.
5171.1 --> 5173.2: The more people that have access to that information,
5173.3 --> 5175.5: and in data brokerage marketplaces, this is true.
5176.1 --> 5179.3: So the freshness of the data is this quality
5180.2 --> 5183.8: that needs to be ensured, and it's my opinion
5183.8 --> 5186.2: that if you explicitly make it known
5186.2 --> 5190.8: that this information has never been exposed
5190.8 --> 5192.1: to a surveillance system,
5192.5 --> 5194.4: then its freshness and its quality
5194.4 --> 5197.0: and the uniqueness to the person and the intent,
5198.5 --> 5200.3: it's exponentially more valuable.
5201.2 --> 5203.9: So that's kind of how I would answer that question,

5204.1 --> 5205.9: that we're always creating information.
5206.1 --> 5208.2: Right now we're all observing each other in the room,
5208.3 --> 5210.8: and it's actually really fresh and new,
5210.9 --> 5214.4: and the information we're getting from each other
5214.4 --> 5216.5: is going into our minds,
5216.6 --> 5218.7: and it's being computed straight away,
5218.8 --> 5221.5: and its value is at its highest point right now.
5221.5 --> 5224.6: But six hours down the track
5224.6 --> 5226.3: when we're all talking at the networking event,
5227.1 --> 5229.8: we've all forgotten a bunch of stuff that we've said
here,
5229.9 --> 5233.5: but we know that information was important,
5233.8 --> 5239.4: and we can then relationally trigger it with each other,
5239.5 --> 5241.3: but that information is experienced decay.
5241.7 --> 5243.3: So this already happens in the real world,
5243.8 --> 5249.4: but if you imagine for that with AI and surveillance,
5249.4 --> 5251.6: that's getting more and more observed,
5251.9 --> 5254.5: which means that the information gets more veracity,
5255.5 --> 5256.7: like stronger and stronger and stronger,
5257.3 --> 5259.3: where part of the benefit of being humans
5259.3 --> 5260.5: is we do forget things, right?
5260.8 --> 5262.9: So information experiences decay.
5263.6 --> 5266.3: So let's try and get that as something,
5266.4 --> 5271.0: as an assumption in AI information systems.
5276.8 --> 5278.4: Maybe that's a stupid question,
5278.6 --> 5281.2: but just thinking this through,
5282.3 --> 5285.4: will the data not ultimately end up
5285.4 --> 5288.7: with surveillance mechanisms anyways?
5288.9 --> 5291.5: Like if I'm, for example, using a decentralized ID
5292.2 --> 5294.9: to purchase stuff online, let's say shoes,
5294.9 --> 5295.8: as you said before,
5296.1 --> 5299.3: I will still send my customer data to this provider,
5299.6 --> 5301.0: and they will still know that I'm using it,
5301.0 --> 5302.3: or I purchased these shoes.
5302.8 --> 5303.5: So it's much more,
5303.9 --> 5307.4: it's not as simple as a mapping exercise
5307.4 --> 5308.4: that is right now,
5308.6 --> 5310.7: because right now you can just simply trace it back
5310.7 --> 5311.6: to this individual,
5311.7 --> 5313.4: because this individual shares this data
5313.4 --> 5315.1: with like decentralized providers.
5315.4 --> 5316.7: But it's very much,
5316.9 --> 5319.1: if I have AI agents floating around,
5319.3 --> 5320.8: or like data mapping exercises
5320.8 --> 5323.1: across like different sources,
5323.2 --> 5324.9: I would still come to the same output.

5325.2 --> 5327.9: So I don't see like how this changes
5327.9 --> 5329.6: the quality of the data itself.
5329.6 --> 5331.0: Serial knowledge proofs.
5331.6 --> 5333.6: So if serial knowledge proofs
5333.6 --> 5335.7: is the translation layer of that information,
5336.0 --> 5337.8: the information never exists,
5338.1 --> 5341.2: but it can be consumed by both parties.
5342.0 --> 5345.1: But in a practical way it still exists, right?
5345.1 --> 5346.4: Because I'm still an individual
5346.4 --> 5348.4: that orders shoes to my house.
5348.5 --> 5350.5: Yeah, because you needed that information to exist
5350.5 --> 5352.0: to get the outcome that you needed.
5352.5 --> 5355.6: However, if you start scrolling
5356.1 --> 5357.9: and type shoes into Google,
5359.8 --> 5362.3: that information is being surveilled.
5362.9 --> 5363.1: Yes.
5363.4 --> 5364.9: And it exists,
5365.1 --> 5367.0: and then they start serving you a whole bunch of ads.
5368.3 --> 5370.5: That's the information that's less valuable
5370.5 --> 5375.9: than if I had a genuine intent to buy shoes,
5376.2 --> 5379.7: and then I put that within an expression, an intent,
5380.3 --> 5383.2: and compose it if it's more secure information
5383.2 --> 5384.2: as serial knowledge proof.
5384.3 --> 5385.3: If it doesn't matter,
5385.3 --> 5389.8: if it's okay for the entity to know who I am,
5390.0 --> 5392.6: then that's up to the sovereign choice.
5393.6 --> 5396.5: But yeah, it's the difference between intent-based data
5396.5 --> 5398.3: and surveillance-based data,
5398.5 --> 5401.1: which I think is important to answer that question.
5401.9 --> 5405.1: And then, yeah, the magic of serial knowledge proofs
5406.4 --> 5410.1: being more computationally efficient long-term
5411.8 --> 5413.8: is the secret sauce here.
5413.8 --> 5413.8: Yeah.
5415.0 --> 5416.4: I agree with you,
5416.5 --> 5418.4: but the ultimate outcome is still the same.
5418.8 --> 5421.9: I still need to share my private data with the vendor
5421.9 --> 5423.9: for them to be able to send me the shoes.
5424.1 --> 5426.2: So the vendor will still be able
5426.2 --> 5428.5: to send me targeted advertising.
5429.9 --> 5433.0: But if I don't consume targeted advertising?
5433.1 --> 5436.1: But in order to actually have a business relationship,
5436.4 --> 5438.5: I still need to share my data, right?
5438.7 --> 5441.7: Because otherwise they can't send me the shoes.
5441.7 --> 5444.9: That's what I'm just getting to ultimately.
5445.2 --> 5448.0: It's like, I agree with you that all the steps in between,

5448.3 --> 5450.9: no one will know which pictures I looked at.
5451.0 --> 5453.6: No one will know what I was actually searching for on Google,
5453.8 --> 5458.6: but they will know which shoes I purchased, right?
5458.8 --> 5460.3: So it's more targeted advertising.
5460.6 --> 5462.3: This is just like, yeah.
5462.6 --> 5463.6: No, I hear you,
5463.7 --> 5467.7: but if you maintain separation of that information,
5468.5 --> 5473.9: so the address that receives the shoes in the mail
5473.9 --> 5477.0: is separate from the intent to buy the shoes,
5477.1 --> 5479.9: which is separate from the interaction
5479.9 --> 5481.8: of the scrolling on the website,
5483.7 --> 5486.2: then that information is in multiple places,
5486.6 --> 5490.1: not observed in one surveilled substrate.
5490.6 --> 5493.0: So that information doesn't become multiplicative
5493.0 --> 5494.7: on creating a persona review.
5495.3 --> 5497.3: It just becomes what is required
5498.1 --> 5501.8: to execute the relationship or the promises
5501.8 --> 5503.9: that are made between the two parties.
5505.3 --> 5507.4: But this would also mean that my address
5507.4 --> 5509.2: is not to attach me as a person.
5510.1 --> 5511.0: What do you mean?
5511.6 --> 5514.5: So basically if I say, okay, like my, as you said,
5514.7 --> 5518.7: delivery is separated from the actual purchase of the shoe.
5519.5 --> 5524.0: The vendor of the shoes still needs to know my address
5524.0 --> 5526.6: and my name to be able to deliver it to my house.
5527.5 --> 5529.5: Or maybe I'm just missing something here.
5529.8 --> 5530.7: You could share a proof
5530.7 --> 5532.7: and then only the delivery driver
5533.2 --> 5536.2: has the proof of that information
5536.2 --> 5539.2: that you could have some sort of expiry or whatever.
5542.1 --> 5542.3: Okay.
5544.7 --> 5546.4: No, yeah, totally, totally.
5546.6 --> 5549.1: But again, yeah, the question is that comes back
5549.1 --> 5551.1: to Sandra's point of convenience
5551.1 --> 5553.1: and operationalization, right?
5553.6 --> 5554.0: I'm just.
5554.2 --> 5555.9: No, no, it's a great point.
5556.9 --> 5557.6: It's okay.
5558.5 --> 5559.4: Yeah, great.
5560.3 --> 5563.0: It's a great point that you're making.
5563.1 --> 5564.0: I 100% agree.
5564.2 --> 5566.7: I'm just thinking about the practical outcomes here, right?
5567.2 --> 5567.6: No, no, no.

5567.7 --> 5570.2: And this is why we're here.
5570.6 --> 5571.1: So thank you.
5571.6 --> 5573.4: We are at break right now
5574.4 --> 5577.5: and we've gone a little bit over the break.
5577.7 --> 5580.9: So let's have like five, 10 minutes to hang out
5580.9 --> 5584.8: and then I'll continue with the PQC conversation.
5585.8 --> 5589.9: So that's the proof of personhood conversation completed.
5590.2 --> 5592.6: The next one is on post-quantum cryptography
5592.6 --> 5595.9: and the re-announcement from Professor Shinichiro Matsuo
5596.4 --> 5599.2: about the Japanese government's initiative
5599.2 --> 5602.2: to do a PQC competition.
5603.0 --> 5605.3: And then we'll share some of the insights about that.
5605.7 --> 5608.6: But if this group then after hearing that information
5608.6 --> 5610.7: is more interested in discussing other things,
5610.7 --> 5612.0: we can discuss other things.
5612.6 --> 5613.9: But yeah, thank you.
5617.9 --> 5618.8: Five, 10 minutes.
5641.5 --> 5644.1: Can I send it to you in the next slide?
5646.2 --> 5646.5: Sure.
5648.1 --> 5649.4: I can put it in the chat.
5650.4 --> 5651.4: Or are you in the Zoom?
5652.4 --> 5653.3: I'm not, no.
5655.8 --> 5657.0: Is everybody here in the Zoom?
5657.2 --> 5659.5: Because it would be best for people in the room.
5660.1 --> 5660.9: Oh, true, okay.
5662.9 --> 5664.6: Maybe we just write it on a piece of paper
5664.6 --> 5665.2: and pass it around.
5665.8 --> 5667.2: If you like, I'll send it to you.
5668.6 --> 5670.4: Could you just write it on a piece of paper?
5670.4 --> 5670.8: Okay.
5671.8 --> 5672.5: Terrible governance.
5673.2 --> 5673.6: All right.
5674.7 --> 5675.3: Let's go.
5678.2 --> 5679.4: Hold on, wait a minute.
5680.4 --> 5681.0: Two questions.
5681.2 --> 5681.9: So I'm right with it.
5683.4 --> 5684.8: So you want to write it on a piece of paper?
5685.0 --> 5685.2: No.
5685.4 --> 5686.8: I think it's a great idea.
5687.1 --> 5687.8: Yeah, I know.
5688.0 --> 5688.9: And then you can tear it up.
5689.2 --> 5689.3: Yeah, yeah, yeah.
5689.3 --> 5690.0: Yeah, you can put it.
5690.4 --> 5691.8: Can you just write it on a piece of paper?
5691.8 --> 5693.0: Yeah, I can put it on a piece of paper.

5693.0 --> 5694.5: I can put it on a piece of paper.
5694.6 --> 5694.7: I can put it on a piece of paper.
5694.7 --> 5696.0: Yeah, I can write it on a piece of paper.
5696.4 --> 5697.3: Well, that would be good.
5697.4 --> 5697.8: It's not as good.
5698.0 --> 5698.5: It's not as good.
5698.5 --> 5698.6: Okay.
5723.6 --> 5724.1: Estimation.
5728.5 --> 5731.0: I'm a co-chair of the identity key access.
5733.0 --> 5733.4: Yeah.
5738.6 --> 5739.1: Okay.
5739.3 --> 5745.5: So, I've got a couple of handles, a public handle is at privacy manage, but if you want
5745.5 --> 5751.9: to just connect on signal, we can do that as well, or WhatsApp or LinkedIn.
5754.1 --> 5755.3: Just find me on LinkedIn.
5755.8 --> 5756.5: Here, I'll get my code.
5757.3 --> 5759.6: So, you just give me a name of it.
5760.3 --> 5761.1: Mitchell Travers.
5768.5 --> 5774.3: I can write it down.
5775.5 --> 5776.4: Oh, it's up here.
5778.0 --> 5778.4: So.
5782.7 --> 5785.6: I was like, no, I'm going to talk to them.
5836.5 --> 5843.6: And I was like, no, not now, because it's one of the rarest things that, you know, I'm
5993.6 --> 5993.9: going to be able to talk to you.
5993.9 --> 5995.7: So, I'll definitely take that.
5996.4 --> 6001.3: The initiative that we're going to have is the goes back to the concept of Google.
6001.9 --> 6003.5: And they were denied that contract.
6003.5 --> 6010.1: So, we, the thing is that, to be clear, our funding isn't around, I have very good trust
6010.1 --> 6018.2: with the private sector, people, people, they could be government-fostered credentials,
6018.8 --> 6023.6: but I think all these people are free because we've got, you know, full-fledged researchers.
6024.6 --> 6025.9: So, they have to use them.
6026.8 --> 6027.8: Let's talk about that.
6027.9 --> 6028.0: Okay.
6029.2 --> 6029.6: Thanks.
6034.6 --> 6035.9: Okay, so...
6035.9 --> 6037.4: So, I was trying to...
6037.4 --> 6039.1: So, Darren and...
6039.1 --> 6039.4: Yeah, yeah.
6040.0 --> 6042.7: And, both were trying to...
6042.7 --> 6043.9: Okay.
6044.9 --> 6047.5: Sorry, I think I didn't catch that one.

6049.1 --> 6049.8: Okay.
6050.5 --> 6051.5: So, anyway...
6051.5 --> 6052.5: We'll have a lot of stuff going on.
6052.7 --> 6052.8: Yeah.
6053.1 --> 6053.8: I have something to say.
6054.4 --> 6055.6: Actually, can I just give you my web?
6055.9 --> 6056.2: That's great.
6057.0 --> 6057.3: Thank you.
6057.3 --> 6057.7: That's a lot.
6057.8 --> 6059.5: Can I have a reach out?
6060.3 --> 6061.8: I was going to go out and talk to you.
6063.2 --> 6063.9: I'll speak for you.
6064.1 --> 6064.4: I'll do it.
6064.9 --> 6065.2: Here.
6065.3 --> 6065.7: Should I call?
6066.0 --> 6066.2: No.
6067.0 --> 6067.5: It's OK.
6067.7 --> 6067.8: You're fine.
6068.0 --> 6068.1: It's OK.